Ultrafast Transient-Grating Spectroscopy of GaAs

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Abstract

Transient-grating spectroscopy probes carrier relaxation and spatial diffusion through the decay of a photoinduced grating. This report analyzes the supplied GaAs autocorrelation, spectrum, pump–probe and transient-grating traces. Fits with nonlinear fringe calibration yield selected Gaussian pulse-intensity FWHM estimates of (112.89 ± 0.10) fs before and (93.902 ± 0.054) fs after the compressor, above the (83.87 ± 0.24) fs Gaussian transform-limit estimate. These uncertainties are propagated formal one-standard-error values, excluding calibration and model uncertainty; fit-window variation is larger. The diffusivity uncertainty below is the regression slope standard error. Biexponential fits summarize the transient traces over the selected interval, while the slow finite-wave-vector rates give an exploratory ambipolar diffusivity of $(3.71 \pm 0.70) \times 10^{-3}\,\mathrm{m^2\,s^{-1}}$. The pump–probe rate is inconsistent with the transient-grating extrapolation, limiting the physical interpretation of that estimate. The data do not directly measure hot-carrier temperature or electron–phonon coupling.

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1 Introduction

Ultrafast optical pulses create and probe non-equilibrium carriers in semiconductors. Their temporal relaxation and spatial transport can be studied by measuring the change in reflected or diffracted probe light after excitation. In transient-grating (TG) spectroscopy, two pump beams interfere in the sample and create a periodic carrier-density pattern. Diffusion reduces the grating amplitude, and the decay rate depends on the grating wave number.

This experiment uses GaAs and combines pulse characterization with pump–probe and TG measurements. The analysis follows the four tasks in the course tutorial: determine pulse duration and spectral bandwidth; assess whether the data can determine hot-carrier cooling and electron–phonon coupling; fit transient-reflectivity and TG decay times; and estimate recombination, ambipolar diffusivity and mobility. The report first derives the relevant models, then describes the optical arrangement and supplied data, and finally presents the results in that order. A final discussion compares the results with their assumptions and available literature.

2 Theoretical background

2.1 Transient-grating formation and carrier decay

An ultrashort pump pulse creates electron–hole pairs in GaAs. Two pump fields with equal intensity I_0 and wave vectors $\mathbf{k}_1, \mathbf{k}_2$ produce the intensity pattern

$$I(\mathbf{r}) = |E_1 + E_2|^2 \tag{1}$$

$$= |E_1|^2 + |E_2|^2 + E_1 E_2^* + E_1^* E_2 \tag{2}$$

$$= 2I_0 \left[1 + \cos\left((\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r} + \phi\right)\right]. \tag{3}$$

The spatial wave number q describes how rapidly the excitation density varies across the sample. The grating period Λ_{TG} is the distance between neighboring excitation maxima, and $q = 2\pi/\Lambda_{\mathrm{TG}}$. A smaller period gives a larger q, steeper density gradients at a fixed modulation amplitude, and faster diffusion-driven decay. Here q describes a variation in space, whereas the optical frequency describes an oscillation in time. For an angle θ between the beams, $q = |\mathbf{k}_1 - \mathbf{k}_2| = 2k \sin(\theta/2)$ and

$$\Lambda_{\mathrm{TG}} = \frac{2\pi}{q} = \frac{\lambda}{2\sin(\theta/2)}. \tag{4}$$

The two-beam interference and resulting spatial carrier modulation are illustrated in Fig. 1.

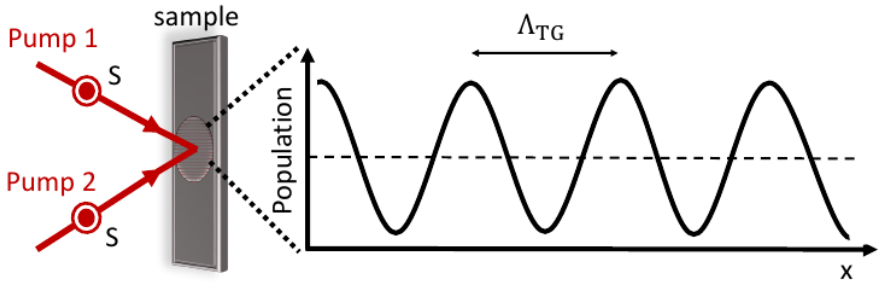


Figure 1: Two crossed pump beams create a sinusoidal excitation pattern in the sample. The grating period Λ_{TG} sets the spatial wave number $q = 2\pi/\Lambda_{\mathrm{TG}}$ used in the diffusion analysis. Reproduced from Fig. 4.5 of the course tutorial.

Write the carrier density as a uniform part plus its sinusoidal grating component,

$$n(x,t) = N_0(t) + N_q(t)\cos(qx). \tag{5}$$

Substitution into the tutorial's one-dimensional diffusion–recombination equation gives

$$\frac{\partial n(x,t)}{\partial t} = D_a \frac{\partial^2 n(x,t)}{\partial x^2} - \frac{n(x,t)}{\tau_{\mathrm{rec}}}. \tag{6}$$

Using $\partial_t n = \dot{N}_0 + \dot{N}_q \cos(qx)$ and $\partial_x^2 n = -q^2 N_q \cos(qx)$, and equating the coefficients of 1 and $\cos(qx)$, respectively,

$$\dot{N}_0 = -\frac{N_0}{\tau_{\mathrm{rec}}}, \qquad \dot{N}_q = -\left(D_a q^2 + \frac{1}{\tau_{\mathrm{rec}}}\right) N_q. \tag{7}$$

$$\int_{N_0(0)}^{N_0(t)} \frac{dN_0}{N_0} = -\int_0^t \frac{dt'}{\tau_{\mathrm{rec}}}, \qquad \int_{N_q(0)}^{N_q(t)} \frac{dN_q}{N_q} = -\int_0^t \left(D_a q^2 + \frac{1}{\tau_{\mathrm{rec}}}\right) dt'. \tag{8}$$

Evaluation yields

$$N_0(t) = N_0(0) e^{-t/\tau_{\mathrm{rec}}}, \tag{9}$$

$$N_q(t) = N_q(0) e^{-(D_a q^2 + 1/\tau_{\mathrm{rec}})t}. \tag{10}$$

For heterodyne detection with a constant local oscillator and $E_{\mathrm{dif}} = \alpha N_q$,

$$|E_{\mathrm{LO}} + E_{\mathrm{dif}}|^2 = |E_{\mathrm{LO}}|^2 + |\alpha|^2 N_q^2 + 2\,\mathrm{Re}(E_{\mathrm{LO}}^* \alpha) N_q, \tag{11}$$

$$S_{\mathrm{het}} \propto N_q \propto e^{-(D_a q^2 + 1/\tau_{\mathrm{rec}})t}. \tag{12}$$

Thus the heterodyne decay rate contains $D_a q^2$, with no factor of two from squaring the grating amplitude. Assuming that the reflectivity change is also linear in the corresponding population, the $q = 0$ transient-reflectivity and finite-q grating traces therefore have the forms

$$S_{\mathrm{pop}}(t) = A_{\mathrm{pop}} e^{-t/\tau_{\mathrm{rec}}} + C_{\mathrm{pop}}, \tag{13}$$

$$S_{\mathrm{TG}}(t) = A_{\mathrm{TG}} e^{-t/\tau_{\mathrm{TG}}} + C_{\mathrm{TG}}, \tag{14}$$

$$\frac{1}{\tau_{\mathrm{TG}}} = \frac{1}{\tau_{\mathrm{rec}}} + D_a q^2. \tag{15}$$

In $1/\tau_{\mathrm{TG}} = 1/\tau_{\mathrm{rec}} + D_a q^2$, the intercept is $1/\tau_{\mathrm{rec}}$ and the slope is D_a.

The provided measurement instructions request two decay components to represent faster carrier relaxation and slower processes such as grating diffusion. We therefore use the phenomenological extension

$$S(t,q) = C + A_f e^{-t/\tau_f} + A_s e^{-t/\tau_s}, \qquad 0 < \tau_f < \tau_s. \tag{16}$$

The assignments of the two fitted components are interpretations of their timescales. To estimate diffusion, the slow-component rate $1/\tau_s$ is regressed against q^2 for the finite-q TG measurements; the single-population diffusion equation above motivates this comparison, but does not by itself derive the biexponential response.

2.2 Ambipolar transport

For ambipolar transport, set the excess electron and hole densities equal, $n = h$, and equate their particle fluxes. With the drift–diffusion expressions

$$\mathbf{j}_e = -D_e \nabla n - \mu_e n \mathbf{E}, \qquad \mathbf{j}_h = -D_h \nabla n + \mu_h n \mathbf{E}, \tag{17}$$

the condition $\mathbf{j}_e = \mathbf{j}_h$ gives

$$-(D_e - D_h)\nabla n = (\mu_e + \mu_h) n\mathbf{E}, \tag{18}$$

$$\mathbf{E} = \frac{D_h - D_e}{\mu_e + \mu_h}\frac{\nabla n}{n}, \tag{19}$$

$$\mathbf{j}_e = -\left[D_e + \mu_e \frac{D_h - D_e}{\mu_e + \mu_h}\right]\nabla n \tag{20}$$

$$= -\frac{D_e\mu_h + D_h\mu_e}{\mu_e + \mu_h}\nabla n. \tag{21}$$

Comparing with Fick's law $\mathbf{j} = -D_a\nabla n$ identifies

$$D_a = \frac{D_e\mu_h + D_h\mu_e}{\mu_e + \mu_h}, \tag{22}$$

$$D_i = \frac{k_B T}{e}\mu_i \quad (i = e, h) \tag{23}$$

$$\Longrightarrow \quad D_a = \frac{2\mu_e\mu_h}{\mu_e + \mu_h}\frac{k_B T}{e} = \mu_a \frac{k_B T}{e}, \tag{24}$$

$$\mu_a = \frac{2\mu_e\mu_h}{\mu_e + \mu_h}. \tag{25}$$

The Einstein step assumes non-degenerate carriers at a common temperature. At high carrier density, many-body and degeneracy effects can change D_a [7].

2.3 Pulse duration and spectral bandwidth

An autocorrelator splits each incoming laser pulse into two *pulse replicas*: copies that travel through separate interferometer arms and are then recombined. Their *optical delay* τ is the difference between their arrival times at the nonlinear crystal. For total propagation distances L_1 and L_2 in air, neglecting dispersion, $\tau \simeq (L_2 - L_1)/c$. Equal travel times give $\tau = 0$ and maximum temporal overlap. Moving one Michelson mirror by Δz changes the round-trip path length by $2\Delta z$, so the delay changes by $\Delta\tau = 2\Delta z/c$ (with the sign set by the chosen arm). Scanning this delay measures how the overlap changes, from which the pulse duration can be inferred under an assumed pulse shape.

For a Gaussian pulse intensity $I(t) = I_0 e^{-t^2/(2\sigma_t^2)}$, the intensity autocorrelation is

$$G(\tau) = \int_{-\infty}^{\infty} I(t)I(t-\tau)\,dt \tag{26}$$

$$= I_0^2 \int_{-\infty}^{\infty} \exp\left[-\frac{t^2 + (t-\tau)^2}{2\sigma_t^2}\right] dt \tag{27}$$

$$= I_0^2 e^{-\tau^2/(4\sigma_t^2)} \int_{-\infty}^{\infty} e^{-(t-\tau/2)^2/\sigma_t^2}\,dt \tag{28}$$

$$= I_0^2\sqrt{\pi}\sigma_t e^{-\tau^2/(4\sigma_t^2)}. \tag{29}$$

Here $t^2 + (t-\tau)^2 = 2(t-\tau/2)^2 + \tau^2/2$ and the substitution $u = (t-\tau/2)/\sigma_t$ reduces the integral to $\sigma_t\int_{-\infty}^{\infty} e^{-u^2}du = \sqrt{\pi}\sigma_t$. The intensity half-maximum satisfies $e^{-t^2/(2\sigma_t^2)} = 1/2$, so $\Delta t_{\text{pulse}} = 2\sqrt{2\ln 2}\,\sigma_t$. The autocorrelation half-maximum satisfies $e^{-\tau^2/(4\sigma_t^2)} = 1/2$, so $\Delta t_{\text{AC}} = 4\sqrt{\ln 2}\,\sigma_t = \sqrt{2}\,\Delta t_{\text{pulse}}$.

The Gaussian time–bandwidth product follows from the field envelope $a(t) = \sqrt{I_0}e^{-t^2/(4\sigma_t^2)}$. Set $F(\nu) = \int_{-\infty}^{\infty} a(t)e^{-i2\pi\nu t}\,dt$ and use $a'(t) = -t\,a(t)/(2\sigma_t^2)$. Differentiating and integrating by

parts on the real axis gives

$$F'(\nu) = -i2\pi \int_{-\infty}^{\infty} t\, a(t) e^{-i2\pi\nu t}\, dt \tag{30}$$

$$= i4\pi\sigma_t^2 \int_{-\infty}^{\infty} a'(t) e^{-i2\pi\nu t}\, dt \tag{31}$$

$$= i4\pi\sigma_t^2 \left([a(t)e^{-i2\pi\nu t}]_{-\infty}^{\infty} + i2\pi\nu F(\nu) \right) \tag{32}$$

$$= -8\pi^2\sigma_t^2 \nu F(\nu), \tag{33}$$

$$\int_{F(0)}^{F(\nu)} \frac{dF}{F} = -8\pi^2\sigma_t^2 \int_0^{\nu} \nu'\, d\nu', \tag{34}$$

$$F(\nu) = F(0) e^{-4\pi^2\sigma_t^2\nu^2}, \qquad F(0) = 2\sqrt{\pi}\sigma_t\sqrt{I_0}, \tag{35}$$

$$|F(\nu)|^2 = 4\pi\sigma_t^2 I_0 e^{-8\pi^2\sigma_t^2\nu^2}. \tag{36}$$

The boundary term vanishes because the Gaussian tends to zero at both limits; $F(0)$ follows by substituting $t = 2\sigma_t u$ in the Gaussian integral. At half maximum, $|F(\nu)|^2 = |F(0)|^2/2$, so

$$8\pi^2\sigma_t^2\nu^2 = \ln 2, \tag{37}$$

$$\Delta\nu = 2|\nu| = \frac{\sqrt{\ln 2}}{\sqrt{2}\pi\sigma_t}. \tag{38}$$

Combining this bandwidth with $\Delta t_{\text{pulse}} = 2\sqrt{2\ln 2}\,\sigma_t$ gives

$$\Delta t_{\text{pulse}} \Delta\nu = 2\sqrt{2\ln 2}\,\sigma_t \frac{\sqrt{\ln 2}}{\sqrt{2}\pi\sigma_t} \tag{39}$$

$$= \frac{2\ln 2}{\pi} \simeq 0.441. \tag{40}$$

From $\nu = c/\lambda$, differentiation gives $d\nu/d\lambda = -c/\lambda^2$; for a narrow spectrum centered at λ_0, $\Delta\nu \simeq (c/\lambda_0^2)\Delta\lambda$.

The optical pulse duration can be measured with a second-order autocorrelator based on second-harmonic generation (SHG). The measured intensity is a function of the relative delay between two pulse replicas. For a Gaussian pulse, the intensity autocorrelation FWHM is $\sqrt{2}$ times the pulse-intensity FWHM. The transform-limited Gaussian time-bandwidth product is $\Delta t \Delta\nu \simeq 0.441$; a larger product can indicate spectral phase/chirp, or measurement and calibration errors. A spectrometer measures wavelength, with $\Delta\nu \simeq c\Delta\lambda/\lambda_0^2$ for a narrow spectrum. This conversion and the time-bandwidth comparison assume that the fitted spectral profile is representative of the pulse spectrum and approximately Gaussian.

2.4 Derivation of the measured interferometric autocorrelation

Interferometric autocorrelation (IAC) measures the integrated second-harmonic signal as a function of the delay between two pulse replicas. This subsection derives that signal, relates its Gaussian overlap widths to the pulse-intensity FWHM, and states the model implemented in `analysis/analyze.py`. The integration variable t is time within a pulse, τ is the optical delay between replicas, and x is the much slower recorded oscilloscope time.

Recorded time is the timestamp at which the oscilloscope samples the signal while the mirror scans. The recorded-time separation $\Delta x = x_{i+1} - x_i$ between neighboring fringe maxima is therefore the elapsed acquisition time between these maxima, expressed here in milliseconds. It differs from the femtosecond-scale optical delay between the pulse replicas. Mirror motion connects the two: a faster change of optical path produces more fringes per unit recorded time, without changing the laser's optical frequency.

2.4.1 Measured signal and expansion of the tutorial formula

Assume identical replicas, an instantaneous second-order response, weak conversion, and phase matching over the relevant bandwidth. The second-harmonic field is proportional to the square of the combined fundamental field. A detector integrating over the pulse therefore records

$$S_{\mathrm{SHG}}(\tau) = \eta \int_{-\infty}^{\infty} \left| [E(t) + E(t-\tau)]^2 \right|^2 dt = \eta \int_{-\infty}^{\infty} |E(t) + E(t-\tau)|^4 dt, \tag{41}$$

where η is a constant conversion and detection factor. This is tutorial Eq. (4.1). For comparison, its printed expansion, Eq. (4.2), is

$$\begin{aligned} I_{\mathrm{AC}}(\tau) = & \int_{-\infty}^{\infty} [I^2(t) + I^2(t-\tau)]\, dt \\ & + \int_{-\infty}^{\infty} [I(t) + I(t-\tau)]\, \mathrm{Re}\{E(t)E^*(t-\tau)\}\, dt \\ & + \int_{-\infty}^{\infty} \mathrm{Re}\{E^2(t)E^{*2}(t-\tau)\}\, dt \\ & + \int_{-\infty}^{\infty} I(t)I(t-\tau)\, dt. \end{aligned} \tag{4.2}$$

Here $I(t) = |E(t)|^2$. Denote the four printed integrals, in order, by T_0, T_1, T_2, T_3. Their relative coefficients must be checked against the defining fourth power. Set

$$a = |E(t)|^2, \qquad b = |E(t-\tau)|^2, \qquad z = E(t)E^*(t-\tau), \qquad zz^* = ab. \tag{42}$$

Then

$$|E(t) + E(t-\tau)|^4 = (a + b + z + z^*)^2 \tag{43}$$

$$= a^2 + b^2 + 2ab + 2(a+b)(z+z^*) + z^2 + z^{*2} + 2zz^* \tag{44}$$

$$= a^2 + b^2 + 4ab + 4(a+b)\,\mathrm{Re}\, z + 2\,\mathrm{Re}(z^2). \tag{45}$$

The identities $z + z^* = 2\,\mathrm{Re}\, z$ and $z^2 + z^{*2} = 2\,\mathrm{Re}(z^2)$ follow by adding conjugate pairs; no real-field assumption is made. Integration gives

$$S_{\mathrm{SHG}} = \eta(T_0 + 4T_1 + 2T_2 + 4T_3). \tag{46}$$

Thus the printed Eq. (4.2) identifies the contributions but omits their relative factors. A single overall normalization cannot correct these omissions.

2.4.2 Evaluation of the Gaussian overlap integrals

Take an unchirped Gaussian field with intensity standard deviation σ_t:

$$E(t) = E_0 e^{-t^2/(4\sigma_t^2)} e^{-i\omega_0 t}, \qquad I(t) = I_0 e^{-t^2/(2\sigma_t^2)}, \qquad I_0 = |E_0|^2. \tag{47}$$

The common Gaussian integral can be evaluated explicitly. Since its integrand is positive,

$$J = \int_{-\infty}^{\infty} e^{-v^2}\, dv > 0, \tag{48}$$

$$J^2 = \int_{\mathbb{R}^2} e^{-(v^2+w^2)}\, dv\, dw = \int_0^{2\pi} d\theta \int_0^{\infty} e^{-\rho^2} \rho\, d\rho \tag{49}$$

$$= 2\pi \left[-\tfrac{1}{2} e^{-\rho^2} \right]_0^{\infty} = \pi, \qquad J = \sqrt{\pi}. \tag{50}$$

For any real shift d, substituting $v = (t-d)/\sigma_t$, so $dt = \sigma_t\, dv$, gives

$$\int_{-\infty}^{\infty} e^{-(t-d)^2/\sigma_t^2} dt = \sigma_t \int_{-\infty}^{\infty} e^{-v^2} dv = \sqrt{\pi}\sigma_t. \tag{51}$$

Define $Q = I_0^2\sqrt{\pi}\sigma_t$. Translation invariance gives the constant term:

$$T_0 = I_0^2 \int_{-\infty}^{\infty} \left[e^{-t^2/\sigma_t^2} + e^{-(t-\tau)^2/\sigma_t^2}\right] dt = 2Q. \tag{52}$$

For the fundamental-frequency term, the carrier phases multiply as

$$z = E(t)E^*(t-\tau) = I_0 e^{-[t^2+(t-\tau)^2]/(4\sigma_t^2)} e^{-i\omega_0 t} e^{i\omega_0(t-\tau)} = I_0 e^{-[t^2+(t-\tau)^2]/(4\sigma_t^2)} e^{-i\omega_0\tau}. \tag{53}$$

Completing the square separately for its two intensity factors gives

$$3t^2 + (t-\tau)^2 = 4(t-\tau/4)^2 + 3\tau^2/4, \tag{54}$$

$$t^2 + 3(t-\tau)^2 = 4(t-3\tau/4)^2 + 3\tau^2/4. \tag{55}$$

Consequently,

$$\int_{-\infty}^{\infty} I(t) z\, dt = I_0^2 e^{-i\omega_0\tau} e^{-3\tau^2/(16\sigma_t^2)} \int_{-\infty}^{\infty} e^{-(t-\tau/4)^2/\sigma_t^2} dt \tag{56}$$

$$= Q e^{-3\tau^2/(16\sigma_t^2)} e^{-i\omega_0\tau}, \tag{57}$$

$$\int_{-\infty}^{\infty} I(t-\tau) z\, dt = I_0^2 e^{-i\omega_0\tau} e^{-3\tau^2/(16\sigma_t^2)} \int_{-\infty}^{\infty} e^{-(t-3\tau/4)^2/\sigma_t^2} dt \tag{58}$$

$$= Q e^{-3\tau^2/(16\sigma_t^2)} e^{-i\omega_0\tau}. \tag{59}$$

Taking the real part of their sum, using $\mathrm{Re}\, e^{-i\theta} = \cos\theta$, yields

$$T_1 = 2Q e^{-3\tau^2/(16\sigma_t^2)} \cos(\omega_0\tau). \tag{60}$$

For the remaining two terms, use

$$t^2 + (t-\tau)^2 = 2(t-\tau/2)^2 + \tau^2/2. \tag{61}$$

The intensity overlap becomes

$$T_3 = I_0^2 \int_{-\infty}^{\infty} e^{-[t^2+(t-\tau)^2]/(2\sigma_t^2)} dt \tag{62}$$

$$= I_0^2 e^{-\tau^2/(4\sigma_t^2)} \int_{-\infty}^{\infty} e^{-(t-\tau/2)^2/\sigma_t^2} dt = Q e^{-\tau^2/(4\sigma_t^2)}. \tag{63}$$

Since $z^2 = I(t)I(t-\tau)e^{-2i\omega_0\tau}$, the second-harmonic overlap is

$$T_2 = \mathrm{Re} \int_{-\infty}^{\infty} z^2 dt = \mathrm{Re}\{e^{-2i\omega_0\tau} T_3\} = Q e^{-\tau^2/(4\sigma_t^2)} \cos(2\omega_0\tau). \tag{64}$$

Inserting all four terms into Eq. (46) gives

$$\begin{aligned} S_{\mathrm{SHG}}(\tau) = \eta Q\big\{2 &+ 8e^{-3\tau^2/(16\sigma_t^2)} \cos(\omega_0\tau) \\ &+ e^{-\tau^2/(4\sigma_t^2)}[4 + 2\cos(2\omega_0\tau)]\big\}. \end{aligned} \tag{65}$$

The cosine terms therefore arise from the relative optical carrier phase, not from an independently assumed oscillation. At $\tau = 0$, the expression gives $16\eta Q$; for separated pulses it tends to $2\eta Q$. Their ratio is the ideal $8:1$ interferometric contrast.

2.4.3 Pulse FWHM versus autocorrelation-envelope width

The pulse duration quoted in this report is the *intensity* FWHM. Its half-maximum condition gives

$$I(t_{1/2})/I_0 = e^{-t_{1/2}^2/(2\sigma_t^2)} = \tfrac{1}{2}, \tag{66}$$

$$t_{1/2}^2 = 2\sigma_t^2 \ln 2, \tag{67}$$

$$\Delta t_{\text{pulse}} = 2|t_{1/2}| = 2\sqrt{2\ln 2}\,\sigma_t. \tag{68}$$

For the two distinct Gaussian overlap factors in Eq. (65),

$$g(\tau) = e^{-\tau^2/(4\sigma_t^2)}, \qquad h(\tau) = e^{-3\tau^2/(16\sigma_t^2)}, \tag{69}$$

$$g(\tau_{1/2}) = \tfrac{1}{2} \Rightarrow |\tau_{1/2}| = 2\sigma_t\sqrt{\ln 2}, \qquad \Delta\tau_g = 4\sigma_t\sqrt{\ln 2} = \sqrt{2}\,\Delta t_{\text{pulse}}, \tag{70}$$

$$h(\tau_{1/2}) = \tfrac{1}{2} \Rightarrow |\tau_{1/2}| = 4\sigma_t\sqrt{\ln 2/3}, \qquad \Delta\tau_h = 8\sigma_t\sqrt{\ln 2/3} = \sqrt{8/3}\,\Delta t_{\text{pulse}}. \tag{71}$$

The two widths are unequal. Only the isolated intensity autocorrelation $T_3 = Qg$ has the familiar $\sqrt{2}$ deconvolution factor. The complete interferometric modulation combines g and h and is not one Gaussian; its envelope FWHM cannot be divided by $\sqrt{2}$ to obtain the pulse duration.

The tutorial's alternative field notation $E(t) \propto e^{-t^2/\Gamma^2}$ corresponds to $\Gamma = 2\sigma_t$. Hence

$$\Delta t_{\text{pulse}} = \Gamma\sqrt{2\ln 2}, \qquad \Delta t_{\text{field}} = 2\Gamma\sqrt{\ln 2}. \tag{72}$$

Thus Γ itself is neither of these FWHM values. The spectral-intensity bandwidth and its time–bandwidth product are treated separately in the preceding subsection.

2.4.4 Recorded-time calibration and the fitted signal

The recorded time x is expressed in milliseconds relative to an initial peak estimate. Let x_0 be the fitted center and $r = x - x_0$. The optical-delay coordinate u, in carrier cycles, is modeled locally by

$$u(r) = fr + b_2 r^2 + b_3 r^3, \qquad \tau(x) = \frac{\lambda_0}{c} u(r), \tag{73}$$

$$\frac{du}{dx} = f + 2b_2 r + 3b_3 r^2, \qquad \omega_0 \tau(x) = 2\pi u(r). \tag{74}$$

Here f, b_2, and b_3 have units of cycles/ms, cycles/ms^2, and cycles/ms^3, respectively. One fundamental fringe cycle corresponds to a delay change λ_0/c. A constant scan speed would give $b_2 = b_3 = 0$; the changing fringe spacing in Fig. 13 motivates allowing a nonlinear mapping. Successive maxima initialize it, and the fitted derivative is checked to be positive throughout the sampled fit window. This is an empirical calibration rather than an independent measurement of mirror displacement.

The blue points in Fig. 13 are calculated from pairs of successive detected maxima in the raw signal. For maxima at recorded times x_i and x_{i+1}, their horizontal coordinate and vertical value are

$$x_{\text{mid},i} = \frac{x_i + x_{i+1}}{2}, \qquad f_i = \frac{1}{x_{i+1} - x_i}. \tag{75}$$

Thus each point represents the inverse fringe spacing, in cycles per millisecond, placed halfway between its two peaks. Closer maxima give a higher value. The code uses SciPy's `find_peaks` function within the fitted window, with a prominence threshold of 5% of that window's raw-signal range and a minimum separation of approximately $0.6/f$, converted to sample indices. Here f is the fitted central fringe rate. The peak positions come from the measured samples, but their selection is therefore not fully independent of the fit. Noise, missed maxima, or extra

detected maxima can affect these estimates; pointwise uncertainties have not been quantified. The red curve instead shows du/dx from the fitted cubic mapping. Agreement supports the delay calibration, but does not independently establish a particular mechanical trajectory.

Write the pulse-intensity standard deviation in optical cycles as $s = c\sigma_t/\lambda_0$. The overlap functions then become

$$g(u) = e^{-u^2/(4s^2)}, \qquad h(u) = e^{-3u^2/(16s^2)}. \tag{76}$$

The implemented fit uses every raw sample within 2.8 ms of the initial peak estimate:

$$\begin{aligned} S_{\mathrm{fit}}(x) = B + A_0 g(u) + h(u)[C_1 \cos(2\pi u) + D_1 \sin(2\pi u)] \\ + g(u)[C_2 \cos(4\pi u) + D_2 \sin(4\pi u)]. \end{aligned} \tag{77}$$

The baseline and harmonic coefficients are independent fit parameters. In the ideal model, with matching signal units and phase origin, they would satisfy

$$B = 2\eta Q, \quad A_0 = 4\eta Q, \quad C_1 = 8\eta Q, \quad C_2 = 2\eta Q, \quad D_1 = D_2 = 0. \tag{78}$$

Allowing them to vary accommodates departures from the ideal recorded amplitudes and phases while retaining the Gaussian overlap widths. This is a phenomenological extension of the ideal SHG model; it does not establish a physical detector-transfer function or remove possible chirp bias.

The fitted width is converted directly to the pulse duration:

$$\sigma_t = s\lambda_0/c, \qquad \boxed{\Delta t_{\mathrm{pulse}} = 2\sqrt{2\ln 2}\, s\lambda_0/c.} \tag{79}$$

No envelope-FWHM deconvolution is used in this conversion. The uncertainty conventions in Sec. 3.4 distinguish formal fit errors from calibration and model uncertainty.

3 Experimental setup and measurement

3.1 Optical arrangement

The tutorial describes a Kerr-lens mode-locked Ti:sapphire femtosecond laser. Its intensity-dependent refractive index produces self-focusing that favors pulsed operation; a prism compressor compensates dispersion and reduces pulse chirp. For autocorrelation, a Michelson interferometer creates two pulse replicas with adjustable delay and a BBO crystal generates a second-harmonic signal proportional to their temporal overlap. For the sample measurement, the beam is split into pump and probe paths; two pump beams cross on the GaAs sample and make the excitation grating. Two probe beams arranged in BOXCAR geometry provide a diffracted signal and a local oscillator. Their interference gives a heterodyne term proportional to the diffracted field, improving sensitivity over direct intensity detection. The tutorial describes phase modulation of the local oscillator with a rotating coverslip and pump modulation with a chopper; a lock-in amplifier extracts the component at the chopper frequency. The overall optical layout is shown in Fig. 2; the individual Michelson autocorrelator and BOXCAR detection principle are shown in Figs. 3 and 4.

3.2 Measurement data and calibration

The supplied data comprise the spectrum file `spektrum.csv`, three before-compressor traces (`vor1`–`vor3`), three after-compressor traces (`nach1`–`nach3`), and five transient files: one pump–probe trace and four TG traces. Each transient file contains delay in its first column and repeated scans in the remaining columns. The measurement instructions specify delay in picoseconds.

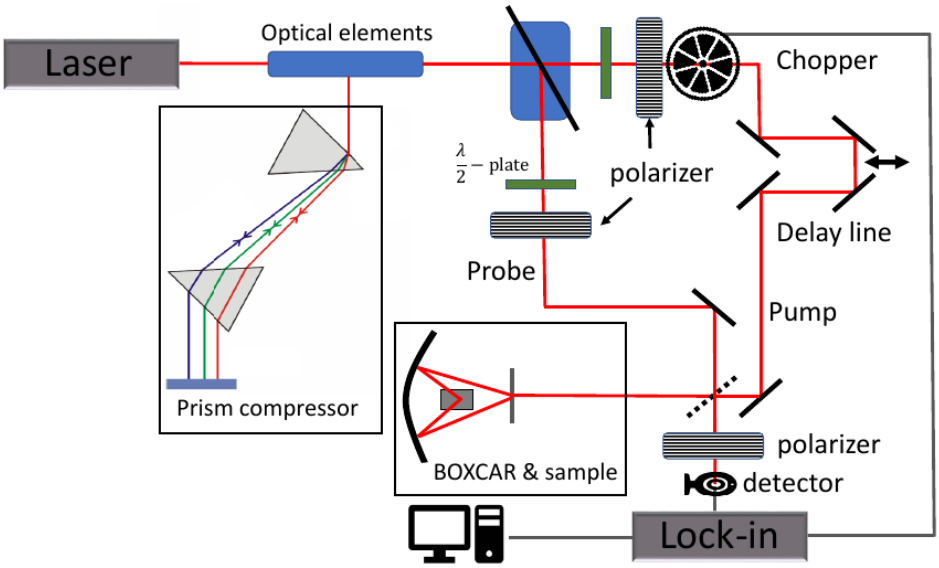


Figure 2: Tutorial schematic of the transient-reflectivity and transient-grating apparatus, including prism compressor, pump and probe paths, delay line, chopper, BOXCAR sample geometry, detector, and lock-in amplifier. Reproduced from Fig. 4.7 of the course tutorial.

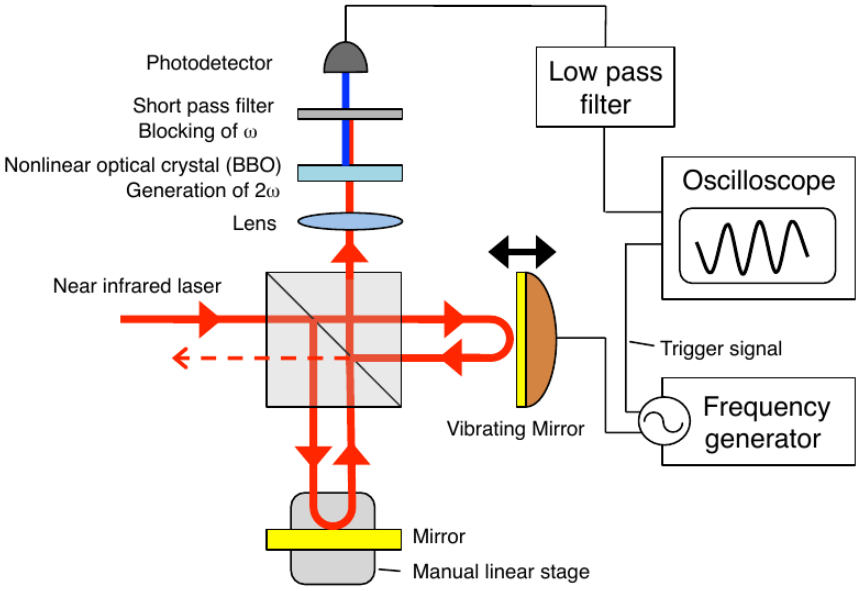


Figure 3: Michelson autocorrelator schematic: the vibrating mirror varies the relative delay between pulse replicas, and a BBO crystal generates the detected second harmonic. Reproduced from Fig. 4.1 of the course tutorial.

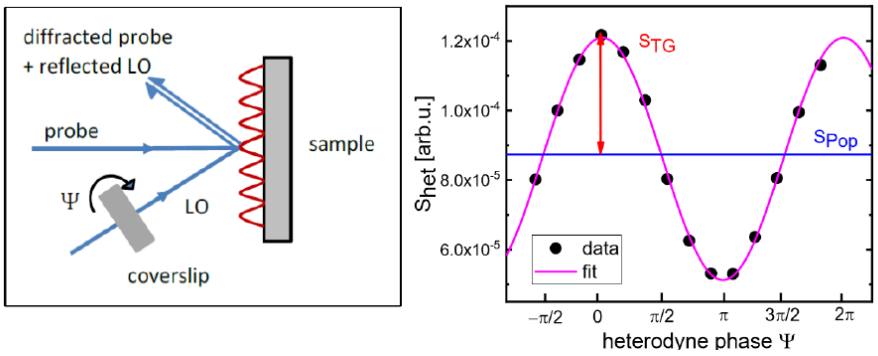


Figure 4: BOXCAR heterodyne detection of the diffracted probe and reflected local oscillator, with the measured signal illustrated as a function of heterodyne phase. Reproduced from Fig. 4.6 of the course tutorial.

The q coordinates are calculated from the filename labels and the supervisor's accompanying email. The numbers 5.6, 6.5, 8, and 12 in the TG filenames specify nominal phase-mask periods in micrometres. The email states that the optical system reduces these periods by a factor of four at the sample, so

$$\Lambda_{\mathrm{sample}} = \frac{\Lambda_{\mathrm{mask}}}{4}, \qquad q = \frac{2\pi}{\Lambda_{\mathrm{sample}}} = \frac{8\pi}{\Lambda_{\mathrm{mask}}}.$$

For example, the nominal 12 µm mask period gives a 3 µm sample period and $q = 2.094\times10^{6}\ \mathrm{m}^{-1}$. All four conversions are listed in Table 2. These are nominal calculated coordinates: mask-period and demagnification calibration uncertainties were not supplied, so their uncertainty remains unquantified. The exact incidence angles and an independent grating-period measurement are also unavailable.

In Fig. 11, the horizontal coordinates therefore come from the nominal grating geometry. The vertical coordinates come from the measured signal columns: retained scans are averaged, their delay dependence is fitted, and the slow rate is calculated as $k_s = 1/\tau_s$. The PP point is assigned $q = 0$ because that measurement has no deliberately imposed periodic excitation grating; its use as a uniform-population reference additionally assumes that transport across the finite probe region is negligible.

The before/after files are compared in Fig. 5. The `vor2` record has a fivefold shorter sampling interval and total duration. Sampling interval alone is not a reason for exclusion, since fringe calibration can accommodate it. The relevant problem is the substantial oscillation remaining at the record boundary, quantified in Part I. The other candidates are ranked by residual RMS divided by the raw-signal range in the fitted window. This chooses a representative fit for display; it does not establish which scan has the smallest physical error.

3.3 Data screening and fit procedure

The analysis was performed with the Python scripts in `analysis/`. For each TG dataset with at least eight repeats, the scan-level deviation score is the root-mean-square of each scan's difference from the pointwise median, standardized by the larger of the pointwise robust scale 1.4826 MAD and its lower-quartile floor. Scores are converted to modified scores using $z_j = 0.6745(s_j - \mathrm{median}(s))/\mathrm{MAD}(s)$; scans above 3.5 are flagged. The primary analysis excludes flagged whole scans and documents their deviations against the retained population; no individual delay points are removed. This is a statistical screen, not proof of acquisition failure, so the effect of retaining flagged scans is also calculated. The screening outcome is described in Sec. 4.3; the flagged scans are compared directly with the retained population in Fig. 10.

The retained scans are averaged pointwise, with standard errors across repeats. The mean traces are fitted by unweighted least squares for $t \geq 0.5\,\mathrm{ps}$ to the ordered biexponential model in Sec. 2.1, using each file through its last recorded delay. The standard errors are plotted but are not fit weights; each recorded delay has equal weight, despite nonuniform delay spacing. The PP record extends to 2000 ps, whereas the TG records end between 26.6 ps and 57 ps; the different windows require a sensitivity check. A 300-replicate bootstrap (Appendix D.1) resamples complete retained scans within each dataset, repeats the averaging and fitting, and propagates slow rates through the finite-q regression. The resulting 95% repeat-scan bootstrap intervals (Appendix D.2) quantify repeat-scan variability conditional on the selected traces and model. They do not include calibration or model uncertainty.

Several experimental conditions needed for full reproducibility were not recorded in the supplied files: measured sample temperature, pump and probe fluence, spot size, sample thickness and orientation, incidence angles, and detector/lock-in settings. The tutorial and `RT` filenames indicate nominal room-temperature measurements; an actual temperature reading is absent. These missing quantities also limit comparison with literature. Numerical results and sensitivity checks are saved in `analysis/results.json` and `analysis/review_checks.json`.

3.4 Uncertainty conventions and propagation

Unless identified as a confidence interval, a quoted ± value denotes one conditional standard error. Nonlinear-fit errors are obtained from the parameter covariance matrix; regression errors are ordinary least-squares standard errors. Bracketed 95% intervals for primary transient lifetimes and diffusivity are percentile intervals from resampling complete retained scans. These quantify different sources of variability and are not interchangeable. Calibration, model, trace-selection and temperature uncertainties remain unquantified and are not included in the quoted statistical errors.

For a derived quantity $y = f(\mathbf{p})$ with covariance matrix Σ, first-order propagation uses

$$u_y^2 = \nabla f^{\mathsf{T}} \Sigma \nabla f. \tag{80}$$

The full spectral covariance is retained for the peak level $A + B$, half-maximum level $A/2 + B$, frequency bandwidth, and transform-limit duration. If s is the fitted pulse width in carrier cycles, then $\Delta t \propto s\lambda_0$ and $\Delta\nu \propto \sigma_\lambda/\lambda_0^2$, so the time–bandwidth product is proportional to $s\sigma_\lambda/\lambda_0$. Its error propagation therefore includes the shared wavelength dependence rather than treating the derived duration and bandwidth as independent. Spectrum and IAC fit parameters from separate records are treated as independent. The common wavelength cancels in the before/after duration ratio. For the nominal intercept lifetime, $u_{1/b} = u_b/b^2$ is a first-order standard error, not a symmetric 95% interval.

Counts, file identifiers, theoretical constants, analysis thresholds and selected fit-window boundaries are specified inputs, not estimated physical results. Recorded axis coordinates are nominal; their instrument-calibration uncertainties are unavailable. Descriptive statistics such as residual RMS and scan scores are identified separately where their uncertainties have not been estimated. No uncertainty of zero is implied by an unavailable uncertainty. Propagation is reproduced by `analysis/report_uncertainties.py`.

4 Results

4.1 Part I: Pulse duration and spectral bandwidth

The pulse durations follow from Eq. (77) with the nonlinear fringe calibration derived above. Figure 7 shows the selected before/after traces. We retain the exclusion of `vor2`: in the last 10% of its record, the RMS signal relative to its whole-record median is 0.397 recorded units, compared with 0.0081 and 0.0058 for `vor1` and `vor3`. Together with Fig. 5, this shows that its

right-hand wing remains appreciably oscillatory at the boundary. These edge RMS values are descriptive diagnostics with unquantified uncertainties. The median is a descriptive reference, not an independently measured baseline. A truncated trace could in principle be fitted, but would constrain the baseline and wings less reliably; we use the better-covered repeats instead.

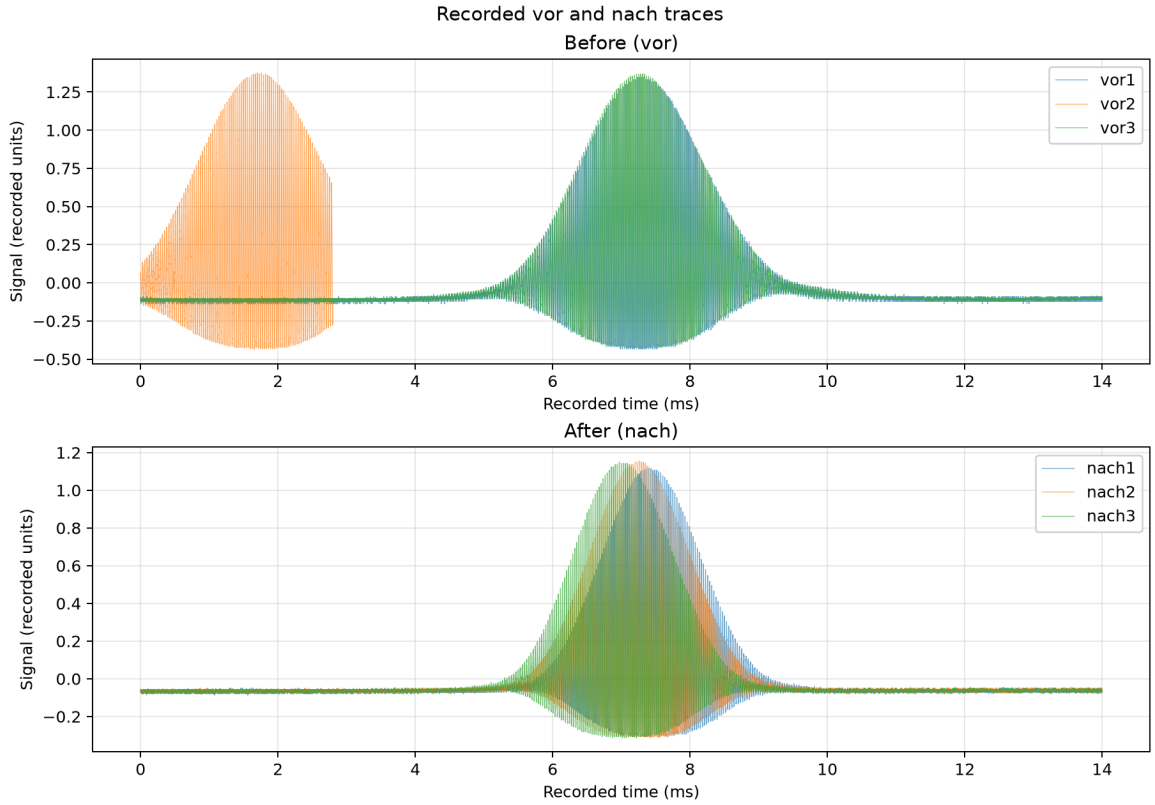


Figure 5: Recorded auxiliary traces grouped as before (vor) and after (nach). The horizontal axis uses recorded time converted to milliseconds.

The measured spectrum and Gaussian fit are shown in Fig. 6. The nominal wavelength coordinates (calibration uncertainty unavailable) span approximately 345 nm to 1041 nm. We fitted all recorded points with

$$S(\lambda) = B + A \exp\left[-\frac{(\lambda - \lambda_0)^2}{2\sigma_\lambda^2}\right]. \tag{81}$$

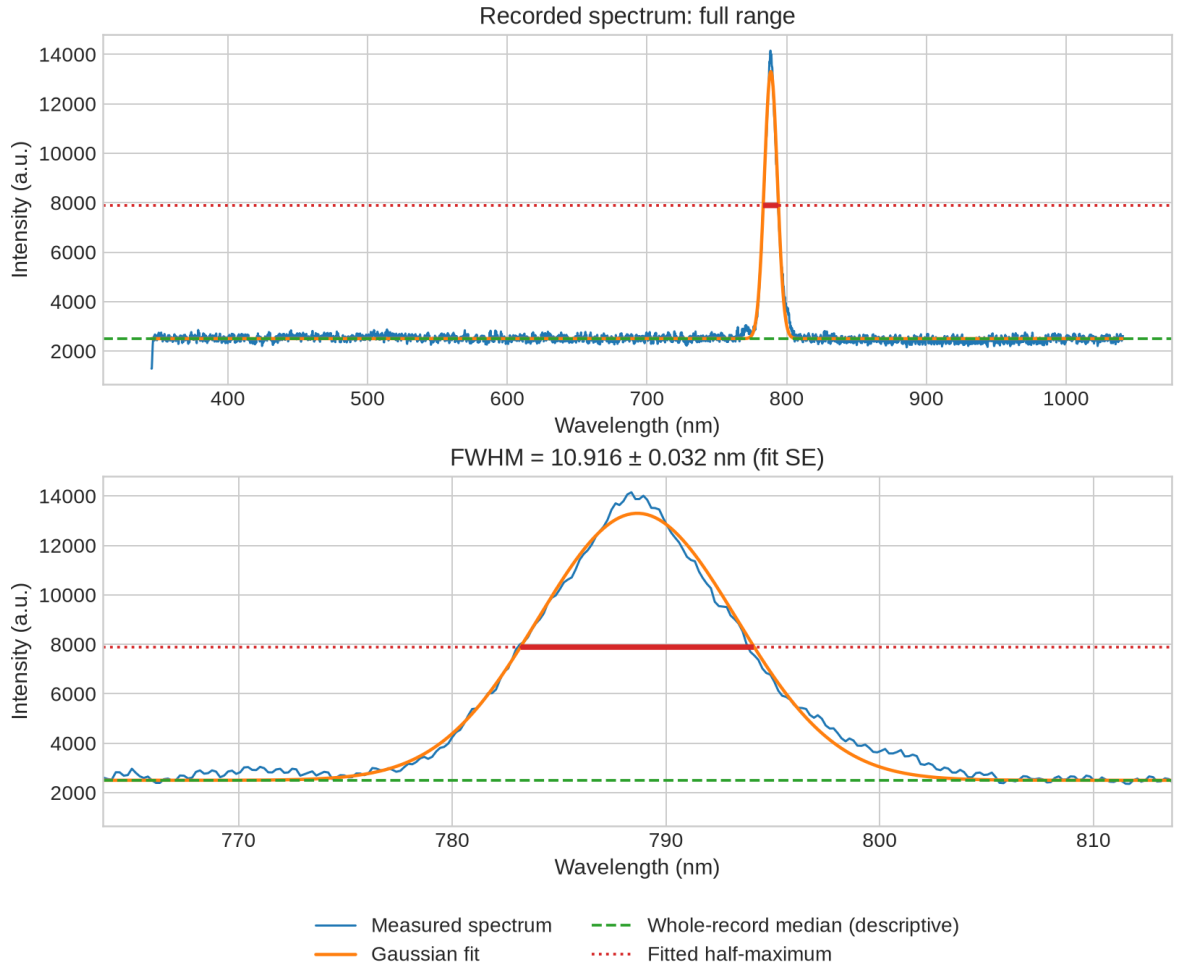
The peak above baseline is A and the fitted peak level is $B + A$. At half peak height above baseline, $S - B = A/2$, so

$$e^{-(\lambda-\lambda_0)^2/(2\sigma_\lambda^2)} = \frac{1}{2}, \tag{82}$$

$$|\lambda - \lambda_0| = \sqrt{2\ln 2}\,|\sigma_\lambda|, \tag{83}$$

$$\Delta\lambda_{\mathrm{FWHM}} = 2\sqrt{2\ln 2}\,|\sigma_\lambda|. \tag{84}$$

The fit gives a centre of (788.656 ± 0.013) nm, peak intensity of (13302 ± 27) a.u., and fitted half-maximum level of (7902 ± 14) a.u., with $\Delta\lambda_{\mathrm{FWHM}} = (10.916 \pm 0.032)$ nm. Propagating the full spectral covariance through $\Delta\nu \simeq c\Delta\lambda/\lambda_0^2$ gives $\Delta\nu = (5.261 \pm 0.015)$ THz and $\Delta t_{\mathrm{TL}} = (2\ln 2/\pi)/\Delta\nu = (83.87 \pm 0.24)$ fs. All quoted errors here are formal one-standard-error uncertainties, excluding calibration and model uncertainty. The median measured intensity is approximately 2501 a u ; its measurement uncertainty is unquantified because repeat spectra and



$$I(\lambda) = B + A\exp[-(\lambda-\lambda_0)^2/(2\sigma_\lambda^2)].$$

Quantity	Estimate	Quantity	Estimate
A (a.u.)	10802 ± 27	B (a.u.)	2500.7 ± 2.5
λ_0 (nm)	788.656 ± 0.013	σ_λ (nm)	4.636 ± 0.014
Peak (a.u.)	13302 ± 27	Half-max. (a.u.)	7902 ± 14
FWHM (nm)	10.916 ± 0.032		

Errors are conditional one-standard-error estimates; calibration and model uncertainties are excluded.

Figure 6: Measured spectrum, Gaussian fit, median level, and fitted half-maximum bandwidth. The displayed bandwidth error is a conditional fit standard error; raw-signal and wavelength-calibration uncertainties are unavailable.

an instrument-noise model are unavailable. The median is a descriptive statistic of all recorded wavelengths, not the fitted background B.

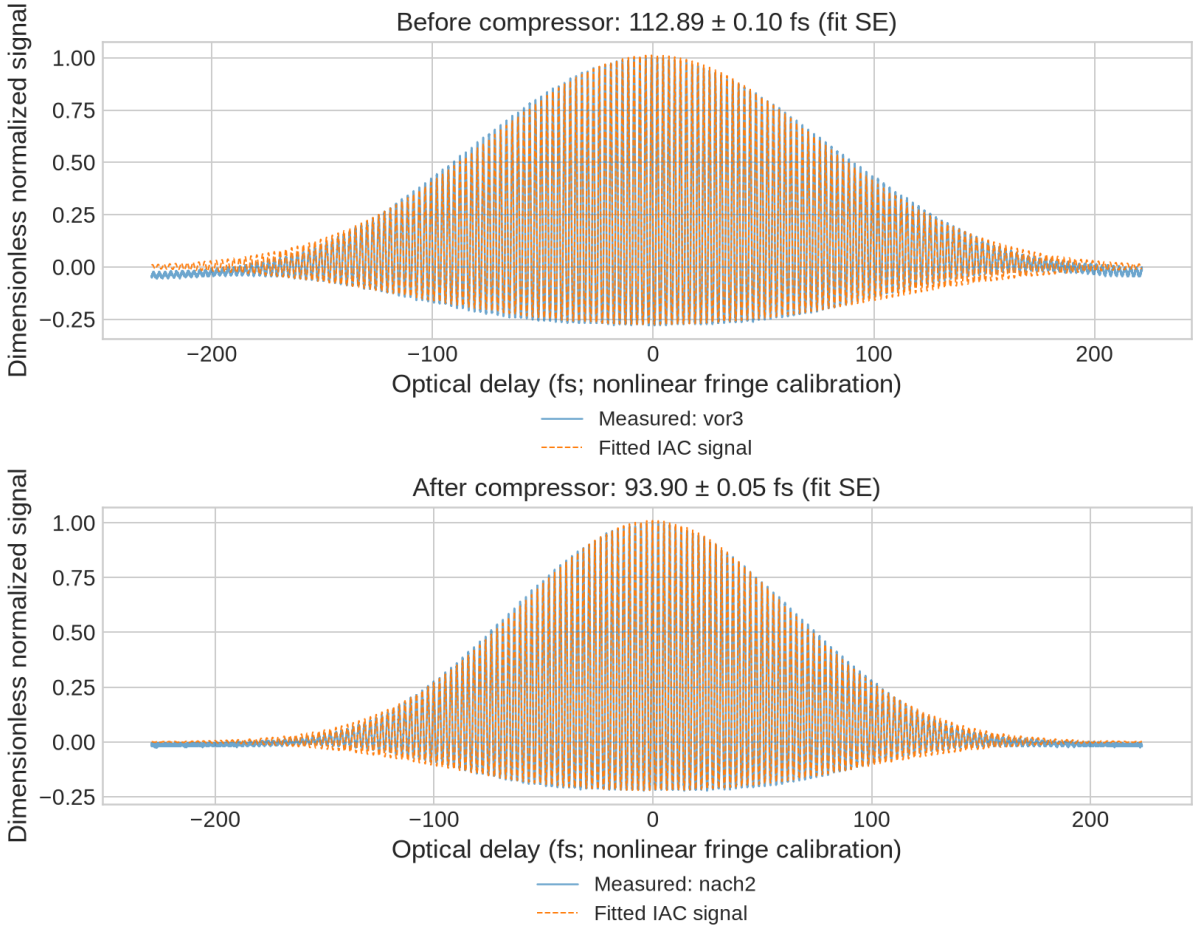
The candidate results are listed in Table 1. The lowest normalized residuals select `vor3` and `nach2`, giving (112.89 ± 0.10) fs and (93.902 ± 0.054) fs. The respective time–bandwidth products are 0.5940 ± 0.0018 and 0.4941 ± 0.0015, both above the exact Gaussian-model limit $2 \ln 2/\pi$. The fitted duration decreases by $(16.823 \pm 0.089)\%$ after the compressor. These are propagated formal standard errors under independent spectrum and IAC fits, with their shared wavelength dependence retained; total model and calibration uncertainties remain unquantified. The change is consistent with compression under these assumptions. These selected-trace estimates are not averages over repeats.

Table 1: Gaussian pulse-intensity FWHM from the raw IAC fits. Errors are conditional covariance standard errors at fixed wavelength, excluding calibration and model uncertainty. Normalized RMS is residual RMS divided by the measured range; its uncertainty has not been estimated.

Trace	Pulse FWHM (fs)	Normalized RMS
vor1	113.09 ± 0.10	0.0230
vor3 (selected)	112.89 ± 0.10	0.0223
nach1	93.99 ± 0.05	0.0135
nach2 (selected)	93.90 ± 0.05	0.0134
nach3	93.95 ± 0.05	0.0135

The changing fringe spacing is visible directly in successive raw-signal maxima (Appendix A, Fig. 13): fitting the same windows with a constant scan speed and the corrected overlap functions gives (74.53 ± 0.39) fs and (71.88 ± 0.28) fs (conditional fit standard errors), with normalized residual RMS 0.107 and 0.080. The cubic calibration lowers those residuals to 0.0223 and 0.0134. These RMS values are descriptive fit diagnostics; uncertainties on the diagnostics have not been estimated. Thus a constant-speed fit can spuriously underestimate pulse duration. This correction was chosen from the observed fringe evolution and fit residuals, not by forcing agreement with the transform limit.

Varying the fit half-window from 2.0 ms to 3.2 ms changes the fitted duration from (111.65 ± 0.08) fs to (114.07 ± 0.10) fs for `vor3`, and from (92.69 ± 0.05) fs to (94.32 ± 0.05) fs for `nach2` (conditional fit standard errors at fixed wavelength). The differences quantify sensitivity to the tested window choices; they are not confidence intervals for the pulse duration. The shifts exceed the conditional covariance errors of 0.10 fs and 0.05 fs. The errors assume independent residuals and an exact model; structured residuals, detector filtering, optical chirp, wavelength calibration, and trace selection are not included. Likewise, restricting the spectral fit to windows of ± 50, ± 30, and ± 20 nm about 789 nm changes the FWHM to (10.783 ± 0.053), (10.651 ± 0.065), and (10.434 ± 0.078) nm, respectively, compared with (10.916 ± 0.032) nm for the full record (conditional fit standard errors). This background/window dependence exceeds the formal 0.03 nm covariance error. These sensitivity ranges are diagnostics, not confidence intervals. The residuals are shown in Appendix A, Fig. 12.



$$
\begin{aligned}
S_{\mathrm{fit}} &= B + A_0 g + h[C_1 \cos(2\pi u) + D_1 \sin(2\pi u)] \\
&\quad + g[C_2 \cos(4\pi u) + D_2 \sin(4\pi u)], \\
g &= e^{-u^2/(4s^2)}, \quad h = e^{-3u^2/(16s^2)}, \\
u &= f(x - x_0) + b_2(x - x_0)^2 + b_3(x - x_0)^3.
\end{aligned}
$$

Parameter	vor3	nach2
B (recorded units)	-0.03824 ± 0.00036	-0.04493 ± 0.00016
A_0 (recorded units)	0.36526 ± 0.00069	0.34501 ± 0.00034
C_1 (recorded units)	-0.23 ± 0.14	0.663 ± 0.027
D_1 (recorded units)	-0.876 ± 0.035	-0.326 ± 0.055
C_2 (recorded units)	-0.131 ± 0.025	0.074 ± 0.017
D_2 (recorded units)	0.080 ± 0.041	-0.106 ± 0.012
x_0 (ms)	0.00869 ± 0.00072	0.00247 ± 0.00038
s (cycles)	18.224 ± 0.016	15.1583 ± 0.0087
f (cycles/ms)	34.05357 ± 0.00031	34.25480 ± 0.00023
b_2 (cycles/ms^2)	-0.1140 ± 0.0010	-0.12715 ± 0.00055
b_3 (cycles/ms^3)	-0.45726 ± 0.00016	-0.45029 ± 0.00019
Pulse FWHM (fs)	112.89 ± 0.10	93.902 ± 0.054

Coefficients refer to the unnormalized recorded signal; the central wavelength is held fixed for the pulse-width errors.

Errors are conditional one-standard-error estimates; calibration and model uncertainties are excluded.

Figure 7: Selected IAC measurements (blue) and fitted signals (orange dashed). Both use the fitted-baseline subtraction and the same measured maximum-deviation normalization. The fit model, parameters, and conditional standard errors are given above.

4.2 Part II: Hot-carrier cooling and electron–phonon coupling

Following the email's assignment of the fast component to relaxation of photoexcited carriers, we report a conditional effective cooling-time estimate $\tau_{\mathrm{cool}}^{\mathrm{eff}} = \tau_f = 0.717\,\mathrm{ps}$, with a 95% repeat-scan bootstrap confidence interval of $[0.698, 0.743]$ ps. Identifying it as a cooling time additionally requires that the fast reflectivity change be proportional to the carrier-temperature excess, with population and optical-response effects separated. That calibration is not supplied. The 0.5 ps fit cutoff excludes the earliest thermalization, but does not by itself exclude picosecond cooling: the tutorial distinguishes roughly 100 fs carrier thermalization (tutorial estimate; uncertainty not supplied) from electron–lattice equilibration over a few picoseconds [3].

The present data therefore support a cooling-time estimate under an optical-response model, but do not uniquely determine a carrier-temperature relaxation time. The systematic PP residuals also make the estimate sensitive to the fitted response: adding a slow contribution changes the fast time to (0.674 ± 0.011) ps (conditional covariance standard error; Sec. 4.4.1). Neither that standard error nor the bootstrap interval above includes the uncertainty in assigning an optical decay component to cooling. The quoted effective cooling time is consequently retained as a conditional interpretation of the baseline fit.

To state exactly what a coupling inference would require, assume constant volumetric heat capacities C_e, C_l and an average electron–phonon coefficient G after excitation:

$$C_e\dot{T}_e = -G(T_e - T_l), \qquad C_l\dot{T}_l = G(T_e - T_l), \tag{85}$$

$$\Delta T = T_e - T_l, \qquad \frac{d\Delta T}{dt} = -G\left(\frac{1}{C_e} + \frac{1}{C_l}\right)\Delta T, \tag{86}$$

$$\int_{\Delta T(0)}^{\Delta T(t)} \frac{d\Delta T'}{\Delta T'} = -G\left(\frac{1}{C_e} + \frac{1}{C_l}\right)\int_0^t dt', \tag{87}$$

$$\Delta T(t) = \Delta T(0)e^{-t/\tau_{\mathrm{cool}}}, \qquad G = \frac{C_eC_l}{C_e + C_l}\frac{1}{\tau_{\mathrm{cool}}}. \tag{88}$$

If $C_l \gg C_e$ and $\tau_{\mathrm{cool}} = \tau_f$, then $G/C_e \simeq 1/\tau_f = 1.394\,\mathrm{ps}^{-1}$, with a conditional bootstrap interval $[1.347, 1.432]$ ps^{-1}. This is a relaxation rate under explicit assumptions, not a measured volumetric coupling coefficient in $\mathrm{W\,m^{-3}\,K^{-1}}$. For nondegenerate electrons of fixed density n, $C_e = \partial(3nk_BT_e/2)/\partial T_e = 3nk_B/2$, but n and the relevant temperature dependence are unknown here. A numerical G is therefore not determined.

The assignment also depends on the fit window: fitting PP only through 33 ps gives $\tau_f = (0.579 \pm 0.010)$ ps, whereas starting the full-record fit at 1 ps gives (0.96 ± 0.13) ps (conditional fit standard errors). These changes exceed the repeat-scan interval. For context, Hunsche et al. report thermalization within 200 fs, while Kash et al. infer an average electron–LO-phonon scattering time near 165 fs below 10^{17} cm^{-3} [4, 6]. The quoted 200 fs is a reported upper bound and 165 fs is an approximate literature value; associated uncertainties are not supplied in the cited abstracts. Neither microscopic time is the same observable as the effective temperature-relaxation time or our reflectivity fit component.

4.3 Part III: Transient-reflectivity and transient-grating decay times

The whole-scan screen was applied to the 44 TG repeats: 12 scans for the nominal mask period 5.6, ten for 6.5, 12 for 8, and ten for 12 (filename period labels). Only scans 4 and 10 in the period-8 dataset exceeded the fixed modified-score threshold of 3.5, with scores of 3.51 and 5.26, respectively; uncertainties on these descriptive scores have not been estimated. Scan numbers count the signal columns starting at one, after the delay column. Both complete scans were excluded, leaving ten period-8 repeats; every scan in the other TG datasets was retained. All five PP repeats were retained because the robust screen requires at least eight repeats. The criterion uses the post-zero fit region and removes entire scans rather than isolated points. Scan

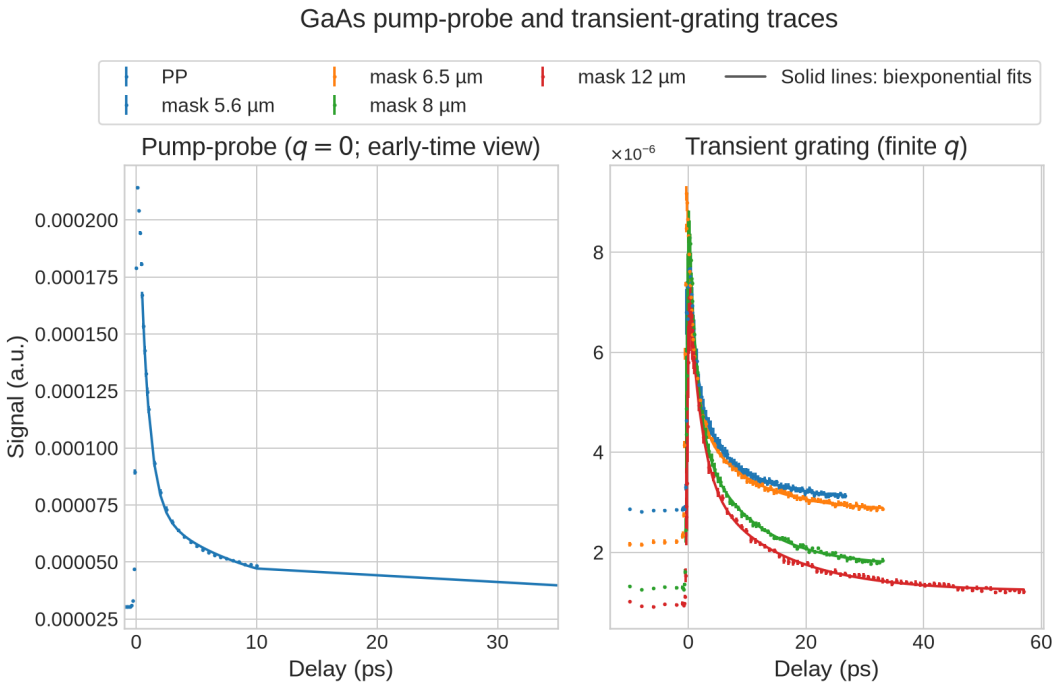
4 is borderline, whereas scan 10 deviates more strongly from the repeat population; their direct comparison with the retained scans is shown in Fig. 10. These statistical deviations do not establish an acquisition fault. The sensitivity of the transport result to retaining the flagged scans is reported in Part IV.

The retained-scan means and biexponential fits are shown in Fig. 8; Fig. 9 shows residuals normalized by the fitted-window signal range. The normalized residual RMS is 1.26% for pump–probe and 0.71–0.81% for TG; uncertainties on these descriptive diagnostics and the scan scores have not been estimated. These small global RMS values do not establish model adequacy: coherent residual structure remains, and the pump–probe plot spans a much longer time interval than the TG data. Each residual panel therefore uses its own delay range. A low RMS alone cannot establish a unique microscopic assignment. In particular, the PP signal continues to decrease after both fitted exponential terms have decayed, whereas the fitted background is constant. This systematic behaviour occurs in all repeats. We retain the PP data as evidence of model inadequacy, but do not use its fitted slow rate to constrain the common TG diffusion regression.

Table 2 gives the fitted fast and slow decay times. The data columns are averaged only after the scan screen; the table gives 95% bootstrap intervals from resampling retained scans, conditional on the selected traces and model. The sample periods use the supplied quarter-period correction, and the first column of each transient file is interpreted in ps. The tabulated periods and wave numbers are nominal calibration values with unquantified uncertainties; $q = 0$ denotes the PP model definition.

Table 2: Biexponential fit lifetimes with 95% repeat-scan bootstrap confidence intervals. Each lifetime is written as the fitted value followed by [lower bound, upper bound], in ps. The bounds are the 2.5th and 97.5th percentiles of fits obtained by resampling complete retained scans. They quantify repeat-scan variability conditional on the selected scans and model, and exclude calibration and model uncertainty. The filename value is the nominal phase-mask period; the sample period is one quarter of that value. Period and wave-number calibration uncertainties are unavailable.

Data	$\Lambda_{\mathrm{sample}}$ (µm)	q ($10^6\,\mathrm{m}^{-1}$)	τ_f (ps)	τ_s (ps)
Pump–probe	–	0	0.717 [0.698, 0.743]	7.878 [7.753, 8.040]
TG 5.6	1.400	4.488	1.206 [1.031, 1.445]	7.205 [6.412, 8.771]
TG 6.5	1.625	3.867	1.298 [1.157, 1.503]	9.200 [8.156, 10.960]
TG 8	2.000	3.142	1.363 [1.064, 1.983]	9.496 [7.933, 12.250]
TG 12	3.000	2.094	2.060 [1.708, 2.463]	13.283 [11.970, 15.170]



$$S_{\mathrm{fit}}(t) = C + A_f e^{-t/\tau_f} + A_s e^{-t/\tau_s}.$$

Trace	τ_f (ps)	τ_s (ps)	A_f	A_s	C
PP	0.717 ± 0.042	7.88 ± 0.79	182.9 ± 6.5	42.3 ± 3.2	35.22 ± 0.28
TG 5.6	1.206 ± 0.056	7.20 ± 0.35	3.294 ± 0.069	2.042 ± 0.076	3.089 ± 0.015
TG 6.5	1.298 ± 0.063	9.20 ± 0.45	3.071 ± 0.062	1.917 ± 0.062	2.838 ± 0.016
TG 8	1.363 ± 0.067	9.50 ± 0.50	4.458 ± 0.093	2.793 ± 0.094	1.729 ± 0.025
TG 12	2.060 ± 0.095	13.28 ± 0.63	3.875 ± 0.085	2.381 ± 0.091	1.225 ± 0.016

A_f, A_s, C are in 10^{-6} recorded signal units. These covariance standard errors differ from the 95% repeat-scan intervals in Table 2. Residuals are $(S_{\mathrm{measured}} - S_{\mathrm{fit}})/(\max S - \min S)$ within the fit window. Errors are conditional one-standard-error estimates; calibration and model uncertainties are excluded.

Figure 8: Mean delay traces over retained scans and matching-color biexponential fits. The PP panel displays the first 35 ps for readability; its fit uses the full record through 2000 ps. Error bars show standard errors across retained scans.

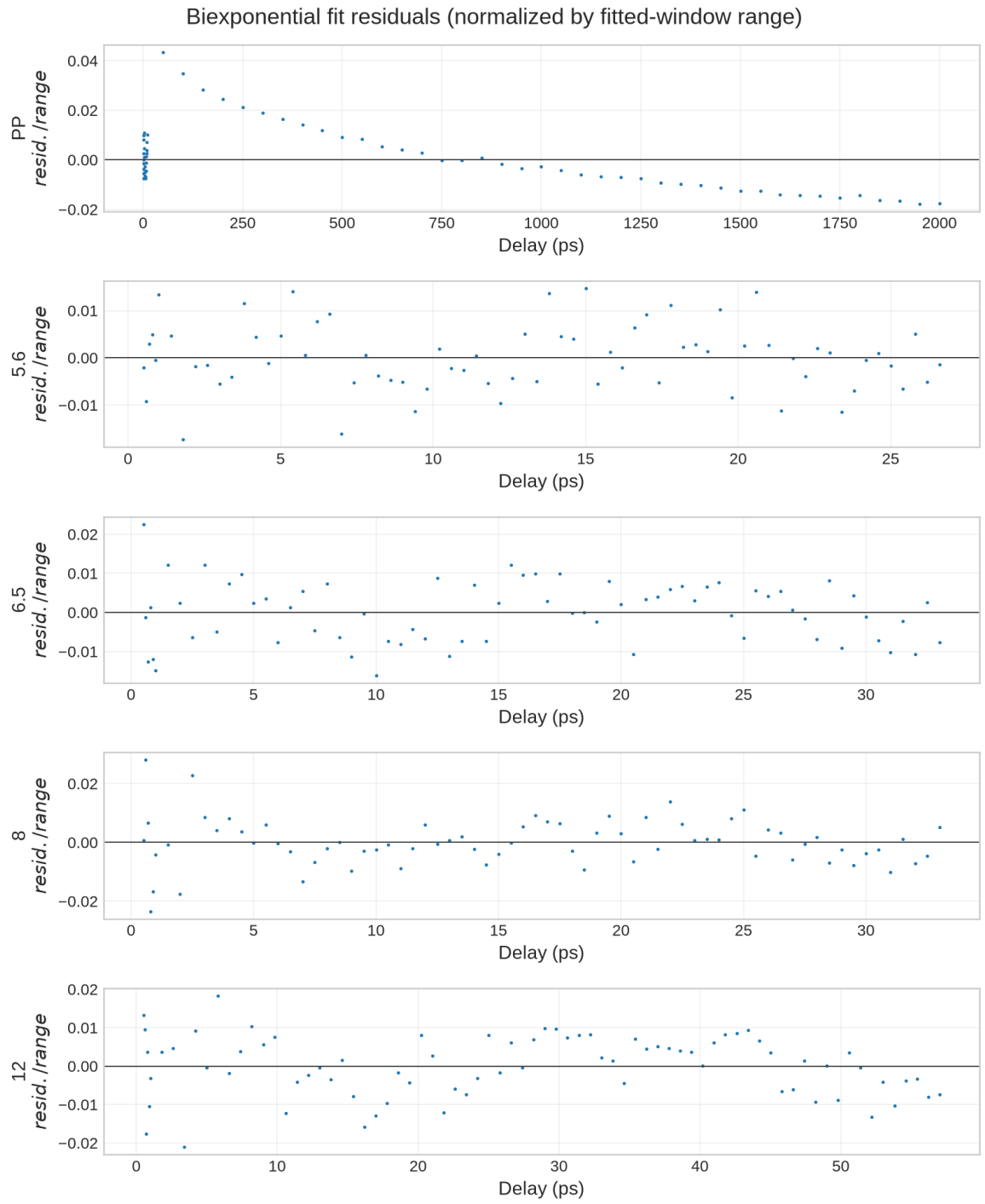


Figure 9: Biexponential fit residuals for the averaged data, normalized by the fitted-window signal range. Each panel has its own delay range, so the long PP record does not compress the TG residuals.

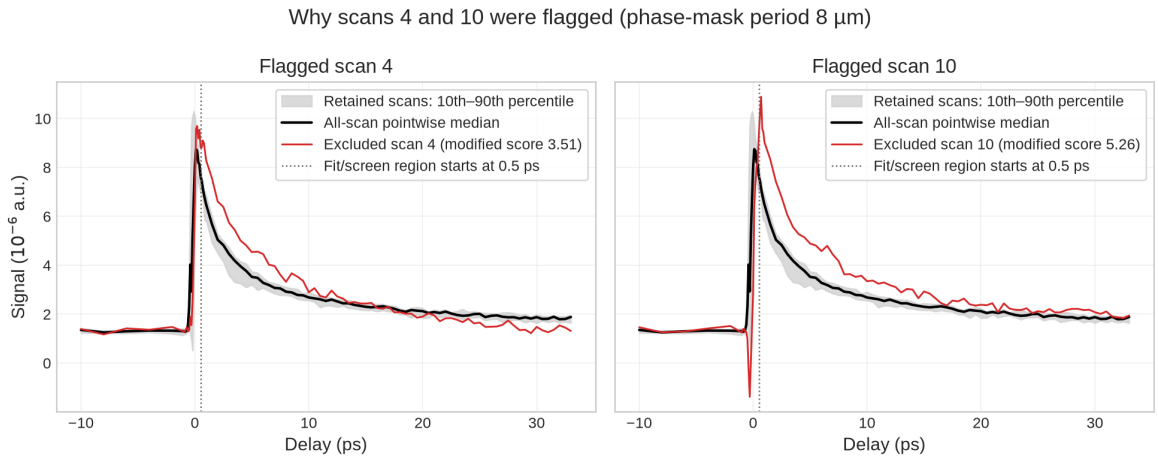


Figure 10: The two statistically flagged scans from the phase-mask-period-8 file, compared with the all-scan pointwise median and the 10th–90th percentile band of retained scans. The vertical dotted line marks the 0.5 ps fit/screen threshold. Scan 4 is borderline; scan 10 shows the larger deviation. Uncertainties on the displayed diagnostic scores have not been estimated.

The fitted lifetimes depend on the observation window. With the PP start held at 0.5 ps, its slow lifetime changes from (2.93 ± 0.11) ps when fitting through 33 ps, to (4.91 ± 0.29) ps through 57 ps, (5.33 ± 0.28) ps through 100 ps, and (7.88 ± 0.79) ps through 2000 ps (conditional covariance standard errors for this comparison). Holding the end at 2000 ps but moving the start to 2 ps gives (3.43 ± 0.19) ps and (375 ± 50) ps (conditional covariance standard errors); the two components then describe different portions of the relaxation. These covariance errors are distinct from the repeat-scan bootstrap intervals in Table 2 and do not account for systematic model mismatch. Consequently, the labels "fast" and "slow" do not identify invariant physical channels across arbitrary fit windows.

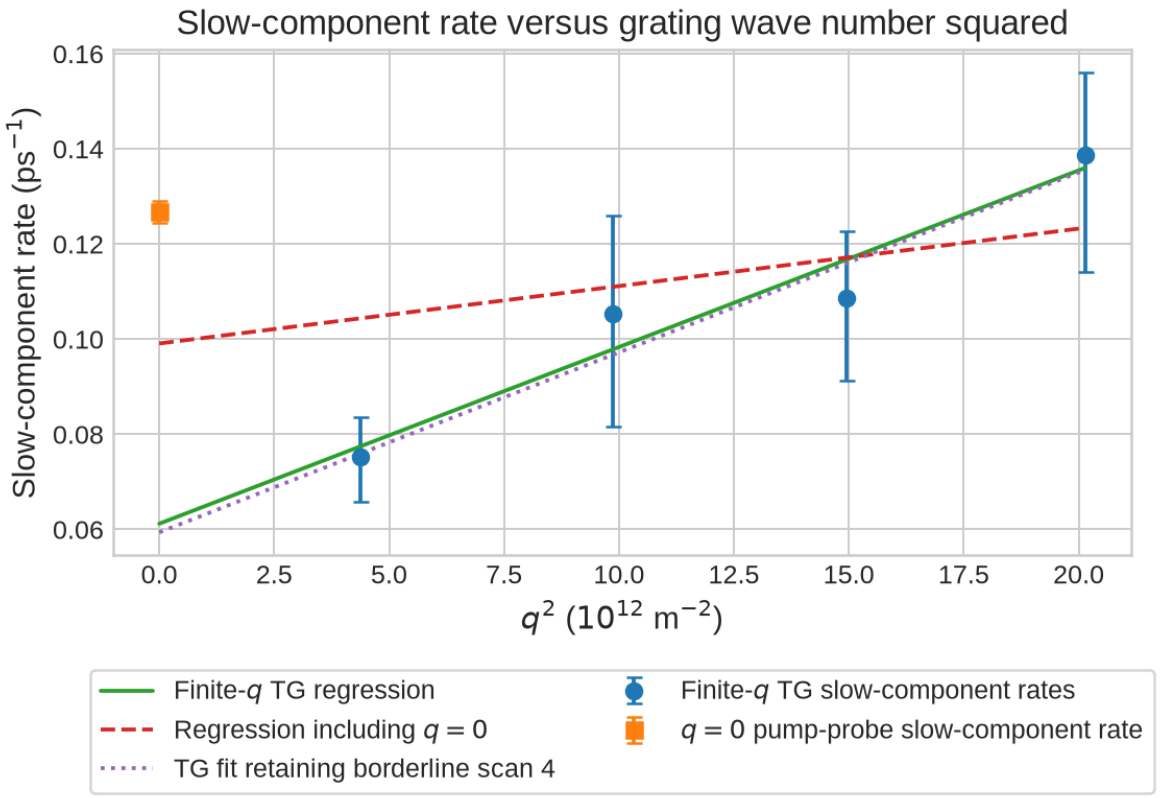
4.4 Part IV: Recombination, ambipolar diffusivity, and mobility

Agreement with the fitted line is summarized by the coefficient of determination R^2, whose definition, interpretation, and limitations are given in Appendix D.3.

For the finite-q TG measurements, the slow-component rates are regressed by unweighted least squares against q^2 as motivated by the diffusion equation. The slope is $(3.71 \pm 0.70) \times 10^{-15}\,\mathrm{m^2\,ps^{-1}}$ ($R^2 = 0.934$; diagnostic uncertainty not estimated), giving an effective ambipolar diffusivity inferred from the slow TG component, $D_a = (3.71 \pm 0.70) \times 10^{-3}\,\mathrm{m^2\,s^{-1}}$, where the stated error is the standard error of the four-point regression slope. Resampling complete retained scans gives a 95% bootstrap interval $[2.31, 4.96] \times 10^{-3}\,\mathrm{m^2\,s^{-1}}$. This interval includes repeat-scan variation but not calibration or model uncertainty. Retaining borderline scan 4 while still excluding scan 10 gives $(3.78 \pm 0.56) \times 10^{-3}\,\mathrm{m^2\,s^{-1}}$ (Fig. 11).

The present data thus permit an effective diffusion estimate under $1/\tau_{\mathrm{TG}} = 1/\tau_{\mathrm{rec}} + D_a q^2$, provided the slow fitted component represents the same population across grating periods. They do not establish a unique material diffusivity: the PP rate disagrees with the TG extrapolation, and the fast and slow TG branches yield different slopes (Sec. 4.4.1). We therefore interpret the value above specifically as the effective diffusivity of the slow fitted TG component. Model and grating-calibration uncertainties remain unquantified beyond the stated statistical errors.

Retaining both flagged scans also gives $D_a = (3.78 \pm 0.57) \times 10^{-3}$ m^2/s. The diffusion slope is therefore insensitive to these two exclusions at the quoted precision. With the primary scan masks held fixed, changing the fit start to 0.2, 1.0, or 2.0 ps gives (3.85 ± 1.02), (4.36 ± 1.25), or $(3.08 \pm 0.42) \times 10^{-3}$ m^2/s, respectively; the common factor 10^{-3} applies to all three values. The



$$k(q) = b + Dq^2, \qquad k = 1/\tau_s.$$

Regression	D (10^{-3} m^2/s)	b (ps^{-1})
Finite-q TG	3.71 ± 0.70	0.0612 ± 0.0095
Including PP	1.2 ± 1.6	0.099 ± 0.019
TG retaining scan 4	3.78 ± 0.56	0.0594 ± 0.0076

Parameter errors are regression standard errors; plotted rate bars are 95% repeat-scan bootstrap intervals. The linear fits use the numerical conversion from ps to s.

Errors are conditional one-standard-error estimates; calibration and model uncertainties are excluded.

Figure 11: Slow-component rates against q^2: horizontal coordinates are calculated from the filename periods and the email's factor-of-four demagnification, while vertical coordinates are inverse slow lifetimes fitted to the recorded traces. Nominal q calibration uncertainties are unavailable. The finite-q line gives the TG-only diffusion estimate; including the $q = 0$ PP rate gives a poor linear fit. Bars show conditional 95% repeat-scan intervals for the rates; the dotted line shows the borderline-scan sensitivity.

uncertainties in these sensitivity regressions are one-standard-error slope estimates, conditional on each selected window and scan set. This window dependence is distinct from the conditional bootstrap uncertainty.

4.4.1 Additional model checks motivated by Chen et al.

Section IV of Chen et al. [2] derives separate signals for grating imaging, total heterodyne detection and direct diffraction. Their weak-excitation total heterodyne expression, Eq. (4), has the structure

$$H(t) \propto F(t)\left[1+\frac{1}{2}e^{-D_a q^2 t}\right], \qquad F(t) = A_r e^{-t/\tau_r} + A_c e^{-t/\tau_c}. \tag{89}$$

The non-diffusing term arises from the population response present in the total detected signal. The paper uses a GaAs/AlAs superlattice and a specific transmission geometry; neither its numerical lifetimes nor its fixed mixing coefficient can be transferred directly to our reflection experiment.

Under the tutorial's interpretation of the supplied TG traces as phase-extracted grating amplitudes, the appropriate quantity is the phase-dependent coefficient, rather than the whole signal in Eq. (89). This distinction follows explicitly by writing

$$S(t,\psi) - B = P(t) + G(t)\cos\psi, \tag{90}$$

$$\frac{S(t,\psi)+S(t,\psi+\pi)}{2} - B = P(t), \tag{91}$$

$$\frac{S(t,\psi)-S(t,\psi+\pi)}{2\cos\psi} = G(t) \quad (\cos\psi \neq 0). \tag{92}$$

The last equation isolates the grating response; it removes the additive population term. The raw phase scans and extraction metadata are unavailable, so this channel interpretation remains an experimental assumption. Adding the paper's non-diffusing term to already extracted TG amplitudes without checking their definition would mix distinct observables.

For a common diffusivity acting on the two components, the isolated response would be

$$G(t) = F(t)e^{-D_a q^2 t} \tag{93}$$

$$= A_r e^{-(1/\tau_r + D_a q^2)t} + A_c e^{-(1/\tau_c + D_a q^2)t}, \tag{94}$$

$$k_r(q) = 1/\tau_r + D_a q^2, \qquad k_c(q) = 1/\tau_c + D_a q^2. \tag{95}$$

Both rate branches must therefore have the same slope against q^2. Applying the existing analysis to both fitted branches gives

$$D_{\text{fast}}^{\text{eff}} = (2.05 \pm 0.63) \times 10^{-2}\ \text{m}^2/\text{s}, \tag{96}$$

$$D_{\text{slow}}^{\text{eff}} = (3.71 \pm 0.70) \times 10^{-3}\ \text{m}^2/\text{s}, \tag{97}$$

where the uncertainties are regression standard errors. These are diagnostic slopes, not two independently established material diffusivities. A paired 300-replicate bootstrap resamples whole retained scans and refits both components together, preserving their correlation. Its conditional 95% interval for $D_{\text{fast}}^{\text{eff}} - D_{\text{slow}}^{\text{eff}}$ is $[0.72, 2.72] \times 10^{-2}$ m^2/s. Thus the simple common-diffusivity assignment is inconsistent with the independently fitted components within the retained-scan variability. Model, fit-window, channel-definition and grating-period uncertainties are excluded. Figure 14 shows the two rate branches. The slow-branch result remains an exploratory effective estimate rather than a validated common transport coefficient.

The paper also emphasizes measuring the population relaxation before constraining grating transport. Our PP residuals show that the existing biexponential-plus-constant model is

inadequate over the complete record. As an empirical diagnostic, we therefore fit

$$S_{\mathrm{PP}}^{(3)}(t) = B + \sum_{j=1}^{3} A_j e^{-t/\tau_j}, \qquad 0 < \tau_1 < \tau_2 < \tau_3, \tag{98}$$

using the same unweighted fit window and retained repeats. The resulting times are (0.674 ± 0.011) ps, (5.22 ± 0.17) ps and (619 ± 60) ps, with conditional covariance standard errors. The normalized residual RMS decreases from 1.26% to 0.303%; uncertainties on these descriptive RMS diagnostics have not been estimated. The extra slow term captures much of the monotonic residual structure, although structured residuals remain (Fig. 14). This does not prove a third carrier process: a non-exponential response or an evolving instrumental background can also produce a slow contribution. In particular, the fitted longest time is not established as a recombination lifetime. The change in the fast fitted time further demonstrates the model dependence of the cooling proxy. The original fit is retained as the stated baseline analysis, with this alternative explicitly shown as a model-sensitivity check. The calculations are reproducible with `analysis/paper_model_comparison.py`.

5 Discussion

The revised autocorrelation analysis gives pulses longer than the spectral Gaussian transform limit, with a shorter fitted duration after compression. Correcting the unequal overlap widths and the nonuniform fringe spacing removes the earlier sub-transform-limit estimates. Pulse durations are obtained directly from the raw-signal fit parameter. Residual structure and fit-window dependence remain, so the numerical durations are effective Gaussian-model estimates. An independently measured mirror trajectory and a phase-sensitive pulse measurement would help separate scan nonlinearity, pulse chirp, and detector response.

The fast PP lifetime can be quoted as a conditional cooling proxy, but the data do not establish the required proportionality between reflectivity and carrier temperature. Carrier thermalization, phonon scattering, energy cooling, and recombination are distinct processes. The cutoff at 0.5 ps limits the earliest dynamics; the stronger reason for caution is the missing response calibration and the observed dependence on fit interval.

The main transport result is the finite-q TG slope shown in Fig. 11. The fit yields

$$D_a = (3.71 \pm 0.70) \times 10^{-3}\,\mathrm{m^2\,s^{-1}} \quad = (37.1 \pm 7.0)\,\mathrm{cm^2\,s^{-1}}.$$

The repeat-scan bootstrap interval is $[2.31, 4.96] \times 10^{-3}\,\mathrm{m^2\,s^{-1}}$. A bulk-GaAs pump–probe study reports an ambipolar diffusivity near $20\,\mathrm{cm^2\,s^{-1}}$ at room temperature (approximate literature value; uncertainty not specified in the cited abstract), so the present estimate is of the same order [5]. This is not a controlled like-for-like comparison because our carrier density, fluence and measured sample temperature are unavailable; density-dependent effects can also change GaAs diffusivity [7].

The finite-q regression intercept corresponds to a nominal recombination time of (16.3 ± 2.5) ps (first-order propagation of the regression intercept standard error). The pump–probe slow lifetime is 7.878 ps with a conditional 95% repeat-scan interval $[7.753, 8.040]$ ps. Equivalently, its rate is 0.1269 $\mathrm{ps^{-1}}$ with a 95% interval $[0.1244, 0.1290]$ $\mathrm{ps^{-1}}$, compared with the TG intercept (0.0612 ± 0.0095) $\mathrm{ps^{-1}}$ (regression standard error); including the $q = 0$ point reduces the linear-fit R^2 to 0.161 (diagnostic uncertainty not estimated). This disagreement is the central limitation of the pump–probe comparison: the $q = 0$ reflectivity signal and finite-q diffracted signal may be sensitive to different mixtures of relaxation processes, while the simple rate law assumes one common population and one recombination intercept. Thus the TG-only slope is an exploratory estimate, and the intercept should not be reported as a measured recombination lifetime.

Here, relaxation processes are physical mechanisms through which the excited sample returns toward equilibrium. Carrier cooling transfers excess carrier energy to the lattice; recombination reduces the number of excited electron–hole pairs; and diffusion redistributes carriers from densely excited regions into less excited regions, reducing the TG modulation without itself removing carriers. These processes can all affect the measured optical response, with different relative contributions to PP reflectivity and TG diffraction. Consequently, the fitted slow PP and TG components may approximate different mixtures of processes. Calling both components "slow" does not identify them with the same recombination mechanism. The two exponential terms are phenomenological descriptions of the traces; fitting them does not establish exactly two independent microscopic relaxation processes or uniquely assign a physical mechanism to either fitted lifetime.

Using $\mu_a = eD_a/(k_BT)$ at an assumed $T = 300\,\mathrm{K}$ gives a conditional mobility of $(1.44 \pm 0.27) \times 10^3\,\mathrm{cm^2\,V^{-1}\,s^{-1}}$. The mobility uncertainty propagates only the regression slope error at fixed assumed temperature; the corresponding 95% repeat-scan interval is $[0.89, 1.92] \times 10^3$ $\mathrm{cm^2/(V\,s)}$. The nominal condition is room temperature, but the actual sample temperature and its uncertainty were not logged, and the Einstein conversion assumes non-degenerate carriers in thermal equilibrium at a common temperature. Therefore this mobility is conditional on assumptions rather than a direct measurement. Across the analysis, the main unresolved uncertainties are the delay and grating-period calibrations, the fit interval and biexponential model, limited grating-period sampling, scan-selection sensitivity, and missing acquisition metadata.

6 Conclusion

The reviewed analysis gives Gaussian pulse-intensity durations of (112.89 ± 0.10) fs before and (93.902 ± 0.054) fs after compression, compared with a (83.87 ± 0.24) fs transform-limit estimate from the spectrum. The quoted pulse and transform-limit errors are formal standard errors; total calibration and model uncertainties remain unquantified. Correct overlap widths and nonlinear fringe calibration are essential to these results. Fit-window effects exceed formal pulse-fit errors. The transient data yield reproducible effective decay constants for a specified window, but do not uniquely identify microscopic cooling or recombination. An effective cooling-time estimate of 0.717 ps (95% repeat-scan interval $[0.698, 0.743]$ ps) can be reported under the assumption of a temperature-sensitive optical response, with additional model and assignment uncertainty unquantified; a volumetric electron–phonon coefficient cannot be obtained without additional heat-capacity and excitation information. The finite-q TG slope gives an effective slow-component diffusivity $D_a = (3.71 \pm 0.70) \times 10^{-3}\ \mathrm{m^2/s}$, with conditional mobility $(1.44 \pm 0.27) \times 10^3\ \mathrm{cm^2/(V\,s)}$ at an assumed 300 K (temperature uncertainty unavailable). The diffusion and mobility errors are regression standard errors and exclude calibration uncertainty. The PP/TG intercept mismatch and measured fit-window sensitivity prevent a firm common recombination-lifetime assignment. The additional checks motivated by Chen et al. expose a slow PP contribution and unequal fast/slow rate slopes, further limiting the common-population transport interpretation (Sec. 4.4.1).

A Autocorrelation calibration and residuals

Figure 12 documents the residual structure of the selected raw-signal fits. The pulse-width analysis in Part I uses these residuals to assess the calibration and fit quality.

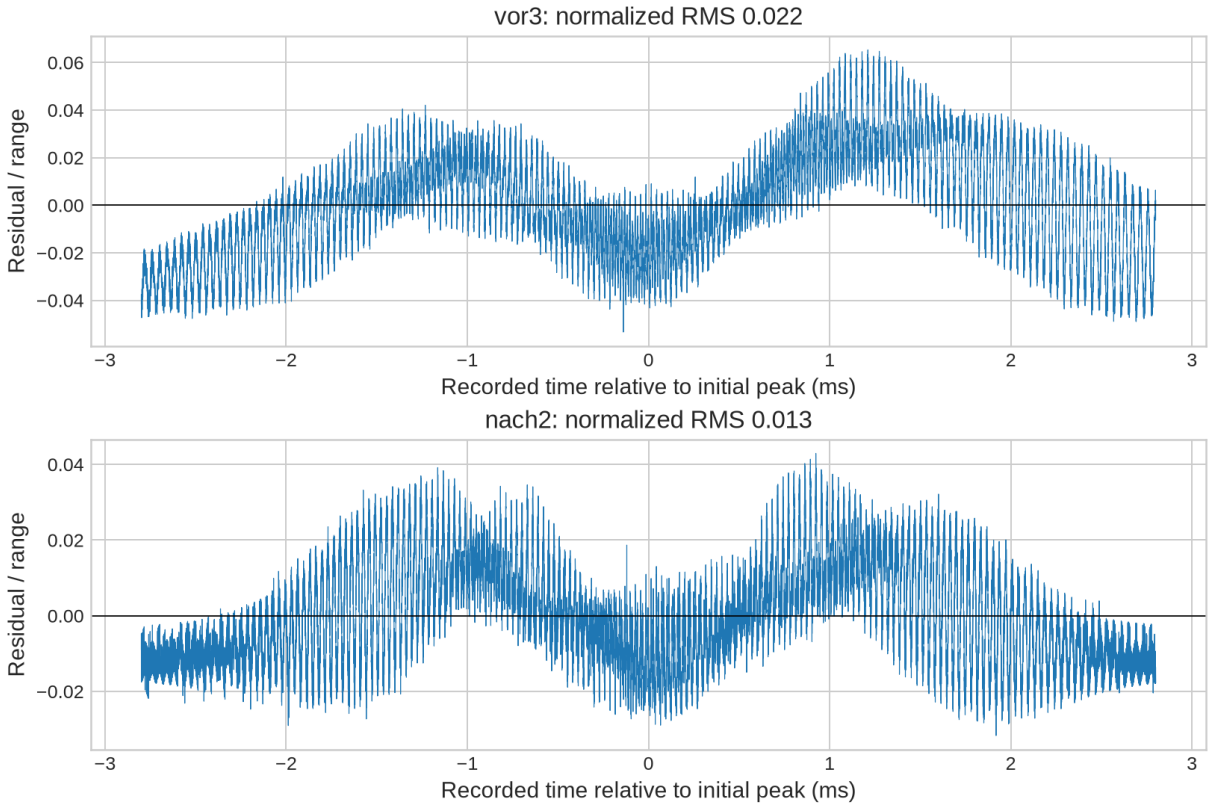
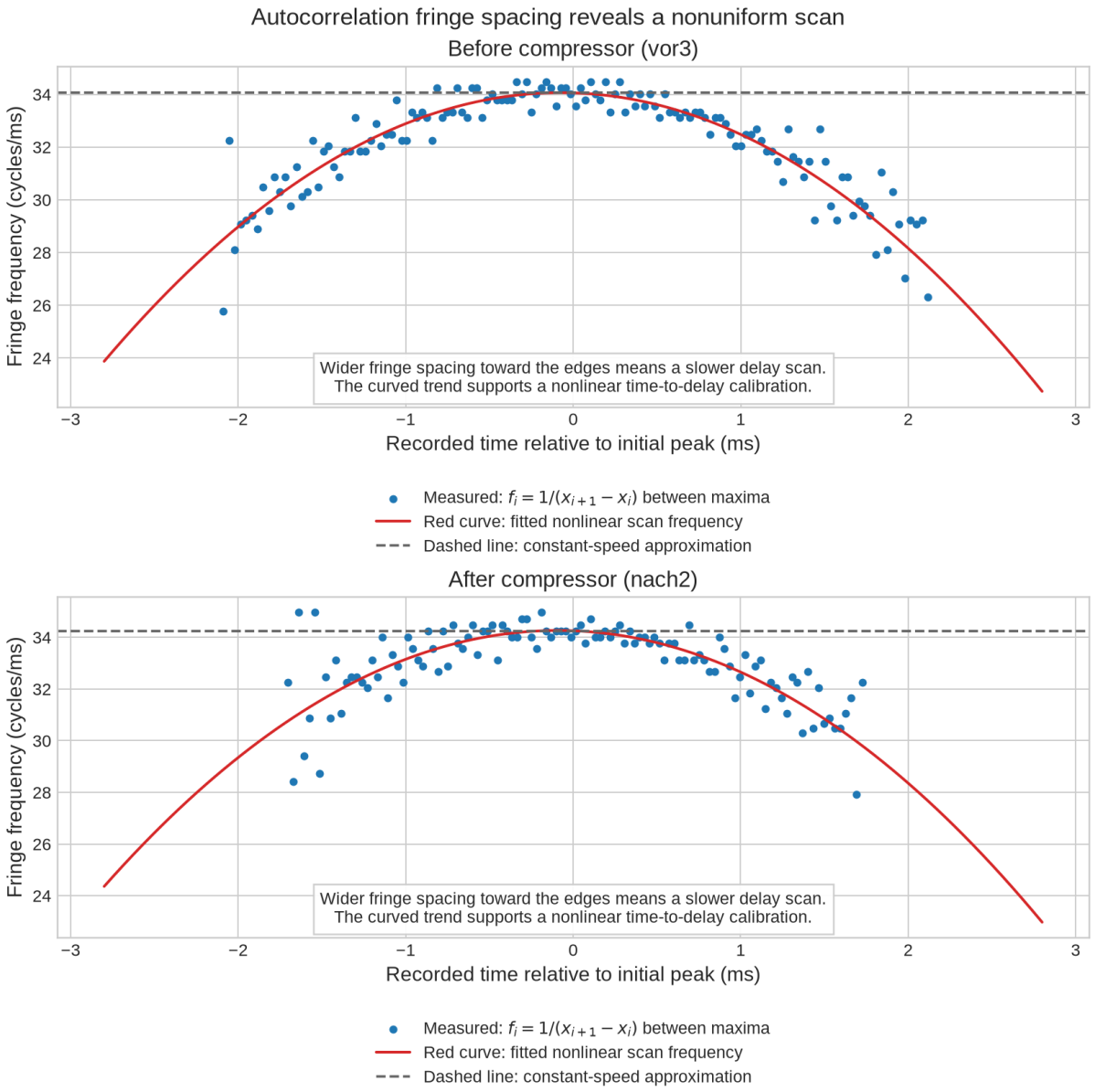


Figure 12: Selected autocorrelation fit residuals, normalized by the measured range in the fitted interval. Structured residuals remain after nonlinear fringe calibration and indicate that covariance errors alone understate uncertainty. Uncertainties on the displayed RMS diagnostics have not been estimated.



$$f_{\mathrm{fit}}(x) = du/dx = f + 2b_2(x - x_0) + 3b_3(x - x_0)^2.$$

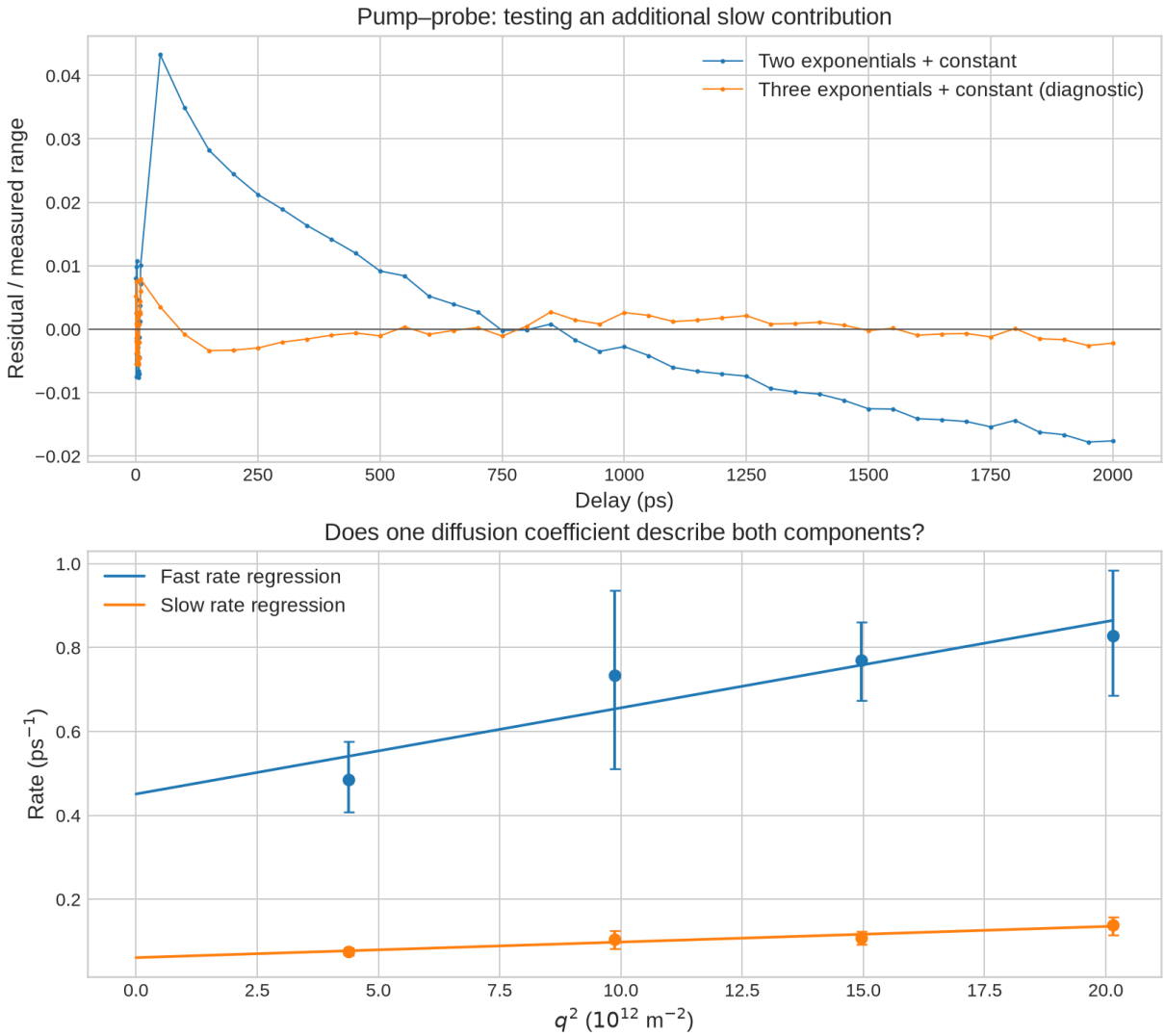
Parameter	vor3	nach2
x_0 (ms)	0.00869 ± 0.00072	0.00247 ± 0.00038
f (cycles/ms)	34.05357 ± 0.00031	34.25480 ± 0.00023
b_2 (cycles/ms^2)	-0.1140 ± 0.0010	-0.12715 ± 0.00055
b_3 (cycles/ms^3)	-0.45726 ± 0.00016	-0.45029 ± 0.00019

The red curves are derivatives of the raw-signal fits, rather than independent fits to the blue points. The dashed lines are f.

Errors are conditional one-standard-error estimates; calibration and model uncertainties are excluded.

Figure 13: Recorded fringe rate: blue points are inverse separations between successive measured maxima, placed at interval midpoints; red curves give the fitted delay-map derivative and dashed lines the central rate. Wider fringe spacing toward the edges supports a nonuniform delay scan. This is a recorded fringe rate, distinct from the optical carrier frequency. Pointwise uncertainties are unquantified; peak selection is explained in Sec. 2.4.

B Population and transport model diagnostics



$$S_m(t) = C + \sum_{j=1}^{m} A_j e^{-t/\tau_j}, \quad m = 2, 3; \qquad k_{f,s}(q) = b_{f,s} + D_{f,s} q^2.$$

Parameter	PP two-exp. / fast rate	PP three-exp. / slow rate
τ_1 (ps)	0.717 ± 0.042	0.674 ± 0.011
τ_2 (ps)	7.88 ± 0.79	5.22 ± 0.17
τ_3 (ps)	–	619 ± 60
A_1	182.9 ± 6.5	181.5 ± 1.8
A_2	42.3 ± 3.2	42.6 ± 1.1
A_3	–	8.47 ± 0.28
C	35.22 ± 0.28	32.82 ± 0.23
D_f, D_s (10^{-3} m^2/s)	20.5 ± 6.3	3.71 ± 0.70
b_f, b_s (ps^{-1})	0.451 ± 0.086	0.0612 ± 0.0095

Amplitudes and offsets are in 10^{-6} recorded signal units. The last two rows give the fast and slow TG regression parameters, respectively. PP errors are covariance standard errors; rate-line errors are regression standard errors. Neither the third PP component nor the rate branches have unique microscopic assignments.

Errors are conditional one-standard-error estimates; calibration and model uncertainties are excluded.

Figure 14: Checks motivated by Chen et al. Top: PP residuals for two and three exponentials plus a constant. Bottom: independent fast- and slow-rate regressions; bars show conditional 95% whole-scan bootstrap intervals. Equal slopes would follow from a shared diffusivity. Wavenumber calibration uncertainty is unavailable.

C Answers to the tutorial questions

This section answers the preparation questions, embedded questions, calculation tasks, and bonus question in the course tutorial [3], in their original order. The headings paraphrase the prompts. The final subsection links the analysis tasks to the experimental results. Numerical constants derived from ideal models are exact expressions or rounded mathematical values, rather than measurements with statistical uncertainties.

C.1 Preparation: optical components and laser operation

What do the optical components do, and how does a solid-state laser work? A prism refracts different wavelengths through different angles and can therefore separate spectral components or provide adjustable dispersion in a compressor. A waveplate introduces a phase difference between two orthogonal polarization components: a half-wave plate rotates linear polarization, while a quarter-wave plate can convert appropriately oriented linear polarization into circular polarization. A polarizer selects one polarization component. The BBO crystal converts part of the fundamental light into its second harmonic.

LASER stands for *Light Amplification by Stimulated Emission of Radiation*. A solid-state laser contains a pumped gain medium, a resonator with mirrors, and a partially transmitting output coupler. Pump absorption establishes a population inversion on the laser transition. A resonant photon can stimulate an excited ion to emit into the same optical mode; repeated passes amplify the light when the round-trip gain exceeds the losses. Gain saturation sets the steady output. The broad gain bandwidth of Ti:sapphire supports many longitudinal modes. Their frequencies satisfy the cavity resonance condition; for an ideal empty linear cavity of length L, their separation is $c/(2L)$.

For equally spaced modes with locked phases, the interference is explicitly

$$E(t) = E_0 e^{i\omega_0 t} \sum_{m=0}^{M-1} e^{im\Delta\omega t} = E_0 e^{i\omega_0 t} \frac{1 - e^{iM\Delta\omega t}}{1 - e^{i\Delta\omega t}}, \tag{99}$$

$$|E(t)|^2 = |E_0|^2 \frac{\sin^2(M\Delta\omega t/2)}{\sin^2(\Delta\omega t/2)}. \tag{100}$$

The limiting peak intensity is $M^2|E_0|^2$ at multiples of $2\pi/\Delta\omega$; adding locked modes narrows each peak. In Kerr-lens mode locking, $n = n_0 + n_2 I$ makes the intense beam center focus more strongly. An aperture or the spatial overlap with the pumped gain region then favors intense pulses through lower effective loss. Dispersion compensation helps maintain their short duration.

C.2 Nonlinear optics and the role of BBO

Why is BBO used, and what does a nonlinear optical response mean? At sufficiently strong fields, the induced polarization is not simply proportional to the applied field. Suppressing tensor indices,

$$P = \epsilon_0\big(\chi^{(1)}E + \chi^{(2)}E^2 + \cdots\big), \tag{101}$$

$$E(t) = E_0 \cos\omega t, \tag{102}$$

$$P^{(2)}(t) = \frac{\epsilon_0 \chi^{(2)} E_0^2}{2}\big[1 + \cos(2\omega t)\big]. \tag{103}$$

The quadratic polarization contains an oscillation at twice the input frequency and radiates second-harmonic light. BBO is non-centrosymmetric, has a useful second-order susceptibility, and permits phase matching by crystal orientation. With suitable phase matching, weak conversion, and adequate bandwidth, $E_{2\omega} \propto E_\omega^2$, so $I_{2\omega} \propto I_\omega^2$. The *field* and *intensity* scaling must be distinguished. The crystal and the filter in Fig. 3 let the detector select this nonlinear overlap signal.

C.3 Gaussian and Lorentzian time–bandwidth products

What are the transform-limited constants for the two pulse shapes? Both widths below are intensity FWHM values, and frequency is measured in cycles per second. For the Gaussian field $a(t) = e^{-t^2/\Gamma^2}$, the Gaussian integration derived in the theoretical background gives

$$|a(t)|^2 = e^{-2t^2/\Gamma^2}, \qquad |\widetilde{a}(\nu)|^2 = \pi\Gamma^2 e^{-2\pi^2\Gamma^2\nu^2}, \tag{104}$$

$$\Delta t = \Gamma\sqrt{2\ln 2}, \qquad \Delta\nu = \frac{\sqrt{2\ln 2}}{\pi\Gamma}, \tag{105}$$

$$K_{\mathrm{G}} = \Delta t\,\Delta\nu = \frac{2\ln 2}{\pi} \simeq 0.441. \tag{106}$$

For the conventional *Lorentzian field envelope*, $a(t) = 1/[1 + (t/a_0)^2]$, with $a_0 > 0$, the intensity is a squared Lorentzian. To evaluate its Fourier integral, set $\Omega = 2\pi\nu$:

$$F(\Omega) = \int_{-\infty}^{\infty} \frac{a_0^2 e^{-i\Omega t}}{(t - ia_0)(t + ia_0)}\, dt. \tag{107}$$

For $\Omega > 0$, close the contour in the lower half-plane, where the exponential decays. The contour is clockwise and encloses the pole at $-ia_0$; its semicircle contribution vanishes. Hence

$$\mathrm{Res}_{z=-ia_0} \frac{a_0^2 e^{-i\Omega z}}{(z - ia_0)(z + ia_0)} = \frac{a_0^2 e^{-a_0\Omega}}{-2ia_0}, \tag{108}$$

$$F(\Omega) = -2\pi i \frac{a_0^2 e^{-a_0\Omega}}{-2ia_0} = \pi a_0 e^{-a_0\Omega}. \tag{109}$$

For $\Omega < 0$, closing in the upper half-plane gives $\pi a_0 e^{a_0\Omega}$; at zero, direct integration gives $a_0[\arctan(t/a_0)]_{-\infty}^{\infty} = \pi a_0$. Thus $F(\Omega) = \pi a_0 e^{-a_0|\Omega|}$. The half-maximum conditions then give

$$\left[1 + (t/a_0)^2\right]^{-2} = \tfrac{1}{2} \quad \Rightarrow \quad \Delta t = 2a_0\sqrt{\sqrt{2} - 1}, \tag{110}$$

$$e^{-4\pi a_0|\nu|} = \tfrac{1}{2} \quad \Rightarrow \quad \Delta\nu = \frac{\ln 2}{2\pi a_0}, \tag{111}$$

$$K_{\mathrm{L}} = \frac{\ln 2}{\pi}\sqrt{\sqrt{2} - 1} \simeq 0.142. \tag{112}$$

These conventions agree with the pulse-shape table in Kärtner's lecture notes [1]. Calling the *intensity* Lorentzian would specify a different pulse and would not justify using this constant. The tutorial does not specify this distinction.

C.4 Benefits, limitations, and stretching of ultrashort pulses

What does the time–bandwidth product imply for an experiment? Short pulses resolve fast changes and provide high peak intensity at a given pulse energy. Their necessarily broad spectrum limits spectral selectivity and may excite several transitions. High peak intensity can also cause unwanted nonlinear responses or damage, while finite pump and probe durations broaden the measured temporal response. The relevant resolution is their cross-correlation at the sample.

A transform-limited pulse has no spectral phase curvature. Writing the spectral phase near ω_0 as

$$\phi(\omega) = \phi_0 + \phi_1(\omega - \omega_0) + \tfrac{1}{2}\phi_2(\omega - \omega_0)^2 + \cdots, \tag{113}$$

$$t_g(\omega) = \frac{d\phi}{d\omega} = \phi_1 + \phi_2(\omega - \omega_0) + \cdots \tag{114}$$

shows that nonzero ϕ_2 gives different frequency components different arrival times. Glass, the gain crystal, and other optics can introduce this dispersion. Nonlinear phase modulation and imperfect compression can also alter the pulse. An enlarged measured time–bandwidth product can additionally reflect an incorrect shape assumption or calibration. Figure 6 supplies the bandwidth and Fig. 7 the pulse-duration comparison; neither alone reconstructs spectral phase.

C.5 Autocorrelator alignment and second-harmonic detection

Why must the replicas overlap in space and time, and why detect the second harmonic? The product $E(t)E(t-\tau)$ is appreciable only where both fields are present. Their focused spots must coincide inside BBO, their polarizations must suit the nonlinear interaction, and their path lengths must bring them within a pulse duration of each other. Steering mirrors set spatial overlap; the translation stage sets the mean delay and the vibrating mirror scans it. A mirror displacement Δx changes the Michelson round-trip delay by $2\Delta x/c$.

The time-integrated fundamental signal is

$$S_\omega(\tau) \propto 2\int |E(t)|^2 dt + 2\,\mathrm{Re}\int E(t)E^*(t-\tau)\,dt. \tag{115}$$

Its delay-dependent part is a field autocorrelation. Its Fourier transform gives the spectral intensity, so it cannot by itself determine pulse stretching caused by spectral phase. Second-harmonic detection instead measures

$$S_{2\omega}(\tau) \propto \int |E(t)+E(t-\tau)|^4 dt, \tag{116}$$

which contains the intensity-overlap integral. It converts femtosecond overlap into a slowly recorded signal as the delay is scanned. For identical replicas with ideal coherent overlap, zero delay gives $16\int|E|^4dt$ and separated pulses give $2\int|E|^4dt$, hence the ideal peak-to-background ratio is $8:1$. These are theoretical ratios, not measured contrasts.

C.6 The autocorrelation envelope and the Gaussian width parameter

Why can the envelope determine pulse duration, and what does the fitted Gaussian width mean? For a specified pulse shape and phase model, the overlap factors vary on the pulse-duration scale while the fringes vary on the optical-period scale. The envelope therefore constrains duration without requiring the absolute optical phase. This is a model-dependent inference: an autocorrelation does not uniquely reconstruct an arbitrary pulse. In this experiment the fringes also calibrate the nonlinear delay scan (Fig. 13), and the actual fit uses the complete trace in Eq. (77).

In the tutorial's expression $E(t) = e^{-t^2/\Gamma^2}$, Γ is a Gaussian scale parameter, *not* an FWHM. The two relevant half-maximum equations are

$$|E(t)| = \tfrac{1}{2} \Rightarrow |t| = \Gamma\sqrt{\ln 2} \qquad \Rightarrow \Delta t_{\text{field}} = 2\Gamma\sqrt{\ln 2}, \tag{117}$$

$$I(t) = e^{-2t^2/\Gamma^2} = \tfrac{1}{2} \Rightarrow |t| = \Gamma\sqrt{\ln 2/2} \qquad \Rightarrow \Delta t_{\text{pulse}} = \Gamma\sqrt{2\ln 2}. \tag{118}$$

For the isolated intensity autocorrelation, completing the square gives

$$G(\tau) = \int_{-\infty}^{\infty} e^{-2[t^2+(t-\tau)^2]/\Gamma^2}\,dt \tag{119}$$

$$= e^{-\tau^2/\Gamma^2}\int_{-\infty}^{\infty} e^{-4(t-\tau/2)^2/\Gamma^2}\,dt \tag{120}$$

$$= \frac{\Gamma}{2}e^{-\tau^2/\Gamma^2}\int_{-\infty}^{\infty} e^{-u^2}\,du = \frac{\sqrt{\pi}\Gamma}{2}e^{-\tau^2/\Gamma^2}, \tag{121}$$

where $u = 2(t - \tau/2)/\Gamma$. Consequently,

$$\Delta t_{\mathrm{AC}} = 2\Gamma\sqrt{\ln 2} = \sqrt{2}\,\Delta t_{\mathrm{pulse}}. \tag{122}$$

The photodiode actually records integrated second-harmonic energy versus delay. The full interferometric envelope contains several overlap terms with different widths, as Eq. (65) shows. Dividing its entire envelope FWHM by $\sqrt{2}$ is therefore unjustified; that factor applies to $G(\tau)$. The report quotes the underlying fitted *intensity* pulse FWHM.

C.7 Calculation tasks: diffusion solution and Fourier transformation

Does the proposed density solve the diffusion equation? Define $k_0 = 1/\tau_{\mathrm{rec}}$ and $k_q = k_0 + D_a q^2$. For the proposed density,

$$n(x,t) = n_0[e^{-k_0 t} + \cos(qx)e^{-k_q t}], \tag{123}$$

$$\partial_t n = -n_0[k_0 e^{-k_0 t} + k_q \cos(qx)e^{-k_q t}], \tag{124}$$

$$\partial_x^2 n = -n_0 q^2 \cos(qx)e^{-k_q t}, \tag{125}$$

$$D_a\partial_x^2 n - k_0 n = -n_0[k_0 e^{-k_0 t} + (D_a q^2 + k_0)\cos(qx)e^{-k_q t}] = \partial_t n. \tag{126}$$

It also has the required initial condition $n(x,0) = n_0[1 + \cos(qx)]$.

How does the Fourier transformation produce tutorial Eq. (4.4)? To handle the ideal periodic grating without integrating a non-decaying density over the entire real line, choose a length L containing an integer number of periods. Define

$$\widetilde{n}_k(t) = \frac{1}{L}\int_0^L n(x,t)e^{-ikx}\,dx, \qquad k = \frac{2\pi m}{L}, \quad m \in \mathbb{Z}. \tag{127}$$

Two integrations by parts yield

$$\frac{1}{L}\int_0^L \partial_x^2 n\, e^{-ikx}\,dx = \frac{1}{L}[\partial_x n\, e^{-ikx}]_0^L + \frac{ik}{L}\int_0^L \partial_x n\, e^{-ikx}\,dx \tag{128}$$

$$= \frac{1}{L}[\partial_x n\, e^{-ikx}]_0^L + \frac{ik}{L}[n\, e^{-ikx}]_0^L - \frac{k^2}{L}\int_0^L n\, e^{-ikx}\,dx \tag{129}$$

$$= -k^2\widetilde{n}_k. \tag{130}$$

The boundary terms vanish by periodicity. Transforming both sides of the diffusion equation therefore gives

$$\frac{d\widetilde{n}_k}{dt} = -(D_a k^2 + k_0)\widetilde{n}_k, \tag{131}$$

$$\int_{\widetilde{n}_k(0)}^{\widetilde{n}_k(t)} \frac{d\widetilde{n}_k}{\widetilde{n}_k} = -(D_a k^2 + k_0)\int_0^t dt', \tag{132}$$

$$\widetilde{n}_k(t) = \widetilde{n}_k(0)e^{-(D_a k^2 + k_0)t}. \tag{133}$$

Initially only the coefficients at $k = 0, \pm q$ are nonzero: they are $n_0, n_0/2, n_0/2$. Substitution into the inverse Fourier series recovers the proposed solution above. The same $-k^2$ rule holds for the Fourier integral of a localized density when the boundary terms vanish at infinity.

C.8 Finite wave number, uniform excitation, and diffusion measurements

What are the lifetimes for finite q and for $q = 0$, and how is D_a obtained? The grating-amplitude lifetime is

$$\tau_q = \frac{1}{1/\tau_{\mathrm{rec}} + D_a q^2}, \qquad \tau_{q=0} = \tau_{\mathrm{rec}}. \tag{134}$$

The $q = 0$ component is spatially uniform and has no lateral density gradient in this ideal model. A pump–probe population measurement approximates this case when lateral diffusion out of the probed region is negligible. A real finite spot or a depth-dependent population can introduce additional transport.

Measure TG amplitude decays for several calibrated periods, calculate $q = 2\pi/\Lambda_{\mathrm{sample}}$, and regress the decay rate against q^2. The slope gives D_a and the intercept gives $1/\tau_{\mathrm{rec}}$ if all traces probe the same population. Alternatively, a matched population trace and one TG trace give

$$D_a = \frac{k_{\mathrm{TG}}(q) - k_{\mathrm{pop}}}{q^2}. \tag{135}$$

Two finite wave numbers also suffice algebraically:

$$D_a = \frac{k(q_1) - k(q_2)}{q_1^2 - q_2^2}, \qquad k_0 = k(q_1) - D_a q_1^2, \tag{136}$$

but additional periods allow the assumed linear dependence to be checked. For the supplied data the sample period is one quarter of the mask period. Figure 11 shows why the measured PP rate cannot simply be used as the common intercept.

C.9 Bonus: other optically generated gratings

Which gratings can be created besides a carrier-density grating? Periodic heating can produce a temperature and refractive-index grating. Periodic stress can launch coherent acoustic waves and a strain grating. Spatially varying pump polarization can create spin or polarization gratings even without a modulated total excitation intensity, provided the material and optical selection rules support them. These gratings probe thermal transport, acoustic propagation, or spin dynamics, respectively; they are possibilities rather than additional signals established in this dataset.

C.10 Heterodyne sensitivity and separation of population and grating signals

Why use heterodyne detection, and how are the two contributions separated? The detector measures

$$|E_{\mathrm{LO}} + E_{\mathrm{dif}}|^2 = |E_{\mathrm{LO}}|^2 + |E_{\mathrm{dif}}|^2 + 2\,\mathrm{Re}(E_{\mathrm{LO}}^* E_{\mathrm{dif}}). \tag{137}$$

For a weak diffracted field, mixing with a stronger local oscillator produces a term linear in E_{dif} that can exceed the directly detected $|E_{\mathrm{dif}}|^2$ term. It also retains phase information. Increasing the local oscillator is useful within detector and noise limits; it does not imply unlimited signal-to-noise improvement.

At each pump–probe delay, scan the coverslip angle and fit the phase dependence shown in Fig. 4. In the tutorial's model,

$$S_{\mathrm{Het}}(t, \psi) = B + S_{\mathrm{Pop}}(t) + S_{\mathrm{TG}}(t) \cos\psi, \tag{138}$$

$$S_{\mathrm{Pop}}(t) = \frac{S_{\mathrm{Het}}(t, 0) + S_{\mathrm{Het}}(t, \pi)}{2} - B, \tag{139}$$

$$S_{\mathrm{TG}}(t) = \frac{S_{\mathrm{Het}}(t, 0) - S_{\mathrm{Het}}(t, \pi)}{2}. \tag{140}$$

Here B is a separately established instrumental offset. A full phase scan can fit an unknown phase origin as well. Opposite-phase subtraction isolates the grating term; averaging isolates the phase-independent population contribution. Pump chopping alone removes the unmodulated background but does not generally remove the pump-induced population term.

The tutorial calls single-probe transient reflectivity homodyne and the added-reference TG measurement heterodyne. These names describe its detection arrangement; using a common source for pump and probe does not decide the classification. Indeed,

$$\Delta I_{\mathrm{PP}} = |E_{\mathrm{probe}}|^2\big(|r_0+\delta r|^2 - |r_0|^2\big) \simeq 2|E_{\mathrm{probe}}|^2\,\mathrm{Re}(r_0^*\delta r), \tag{141}$$

so the reflected probe supplies an internal reference for the induced field change. In terminology that reserves heterodyne for a frequency-shifted reference, same-frequency mixing is called homodyne; the tutorial uses the broader spectroscopy convention.

How are recombination and diffusion rates separated experimentally? After extracting the two phase components, fit their time dependence and use $k_{\mathrm{TG}} - k_{\mathrm{Pop}} = D_a q^2$, or vary the grating period and determine slope and intercept as above. If $E_{\mathrm{dif}} \propto n_q$, direct diffracted intensity instead decays as $|n_q|^2 \propto e^{-2k_q t}$; its measured rate must be divided by two before applying the amplitude-rate law. These procedures require matched physical components. The two empirical biexponential times in Table 2 do not by themselves establish two microscopic populations or a unique recombination time.

C.11 Chopper, lock-in amplifier, and prism compressor

How do modulation and compression help the measurement? A rotating slotted chopper periodically blocks the pump. The pump-induced probe response is then modulated at a known reference frequency. A lock-in multiplies the detector signal by synchronized sine and cosine references and averages over many cycles. For $s(t) = A\cos(\Omega t+\varphi)+B$, an averaging interval containing complete cycles gives

$$X = 2\langle s(t)\cos\Omega t\rangle = A\cos\varphi, \tag{142}$$

$$Y = -2\langle s(t)\sin\Omega t\rangle = A\sin\varphi, \tag{143}$$

because $\langle\cos^2\Omega t\rangle = \langle\sin^2\Omega t\rangle = 1/2$ and the mixed and DC-reference averages vanish. Thus $\sqrt{X^2+Y^2} = |A|$. A real chopped signal contains harmonics; the lock-in selects the component at its chosen reference frequency. This suppresses DC backgrounds and noise outside the detection bandwidth, with a tradeoff between averaging time and acquisition speed.

A prism compressor gives different wavelengths adjustable optical path lengths. Properly chosen prism spacing and insertion compensate accumulated dispersion so that spectral components overlap more closely in time. It improves time resolution and peak intensity at fixed pulse energy; it does not require a narrower spectrum. Residual dispersion, losses, and alignment sensitivity limit the achieved compression. The compressor location and modulation arrangement are shown in Fig. 2.

C.12 Ambipolar mobility limits and the band structure

What do the equal- and unequal-mobility limits mean? Starting from the tutorial's expression,

$$\mu_a = \frac{2\mu_e\mu_h}{\mu_e+\mu_h}, \tag{144}$$

$$\mu_e = \mu_h = \mu \Rightarrow \mu_a = \mu, \tag{145}$$

$$\mu_j \gg \mu_i \Rightarrow \mu_a = \frac{2\mu_i}{1+\mu_i/\mu_j} \longrightarrow 2\mu_i. \tag{146}$$

Equal mobilities give the same ambipolar mobility. For strongly unequal mobilities, the internal electric field couples the two species and the slower species limits the joint transport. The printed second limit in tutorial Eq. (4.14) is a typographical error: it should state $\mu_a \simeq 2\mu_i$ for $\mu_j \gg \mu_i$.

Band curvature determines the effective-mass tensor:

$$(m^{*-1})_{\alpha\beta} = \frac{1}{\hbar^2}\frac{\partial^2 E}{\partial k_\alpha \partial k_\beta}. \tag{147}$$

For an isotropic parabolic band in the relaxation-time approximation, the steady drift follows from

$$0 = q_c\mathbf{E} - \frac{m^*\mathbf{v}_d}{\tau_m} \quad \Rightarrow \quad \mu = \frac{|\mathbf{v}_d|}{|\mathbf{E}|} = \frac{e\tau_m}{m^*}, \tag{148}$$

where $q_c = \pm e$ and τ_m is the momentum-relaxation time. Greater curvature means smaller effective mass and, at fixed τ_m, larger mobility. Scattering, band anisotropy, and occupation of different valleys also matter; band curvature alone does not determine the mobility.

C.13 Where the tutorial's analysis tasks are answered

Part I reports the fringe-based delay calibration, spectral bandwidth, pulse widths, and time–bandwidth comparison, with Figs. 6, 7, and 13. Part II gives the conditional cooling-time interpretation and derives the coupling relation; a volumetric electron–phonon coefficient remains undetermined because the necessary heat-capacity and excitation information is missing. Part III gives the transient-reflectivity and TG decay constants with confidence intervals in Table 2. Part IV and the discussion assess the effective diffusion coefficient, conditional mobility, literature comparison, and failure of a common recombination intercept. These answers retain the distinction between measured optical decay times and model-dependent material quantities.

D Statistical methods and interpretation

D.1 Bootstrapping

Bootstrapping estimates the sampling variability of a calculated result by repeatedly resampling the available measurements. Sampling is performed *with replacement*: the same measurement can occur more than once in a resampled dataset, while another may be absent. Each resample contains the same number of measurements as the original dataset. The complete analysis is repeated on every resample, producing a distribution of estimates rather than a single fitted value.

In this report the resampling unit is a complete retained scan. Suppose a file contains n retained scans $S_1(t), \ldots, S_n(t)$, recorded at the same delay coordinates. For replicate r, draw n indices independently and uniformly from $\{1, \ldots, n\}$:

$$J_1^{(r)}, \ldots, J_n^{(r)}, \qquad \overline{S}^{*(r)}(t) = \frac{1}{n} \sum_{j=1}^{n} S_{J_j^{(r)}}(t).$$

Fit this resampled mean with the same biexponential function, fit interval, and numerical procedure as the original mean. Repeat this for $R = 300$ resamples. For a lifetime, the resulting estimates are $\tau^{*(1)}, \ldots, \tau^{*(R)}$. The reported central estimate remains the fit to the original retained-scan mean; it is not replaced by the mean of the bootstrap estimates.

For quantities involving several files, scans are resampled separately within each file. Each replicate then repeats all relevant fits and derived calculations. In particular, slow lifetimes are converted to rates and regressed against the unchanged nominal q^2 coordinates to obtain a bootstrap diffusion estimate. The paired fast/slow diagnostic uses the same resampled scans for both components, preserving their correlation.

Resampling whole scans preserves relationships between delay points within a scan. Treating all individual delay points as independent would discard that structure. The method nevertheless assumes that the retained scans adequately represent repeat measurements under comparable conditions; it cannot recover sources of variability absent from those scans.

D.2 Repeat-scan bootstrap intervals

A *repeat-scan bootstrap interval* summarizes the spread of the estimates obtained by the procedure above. The conditional 95% percentile interval for a parameter θ is

$$\mathrm{CI}_{95\%}(\theta) = \left[Q_{0.025}(\theta^*),\ Q_{0.975}(\theta^*) \right],$$

where Q_p denotes the p quantile of the bootstrap estimates. Its endpoints are the 2.5th and 97.5th percentiles, enclosing the middle 95% of those estimates. The interval need not be symmetric about the original estimate.

The quantiles are calculated by sorting the N bootstrap estimates as $x_0 \le x_1 \le \cdots \le x_{N-1}$ and applying NumPy's default linear interpolation. With indices starting at zero and $0 < p < 1$, define

$$h = (N-1)p, \qquad j = \lfloor h \rfloor, \qquad a = h - j.$$

The quantile is the interpolated value

$$Q_p = (1-a)x_j + a x_{j+1}.$$

An integer h gives $a = 0$ and selects a sorted estimate directly; otherwise the quantile lies between its two neighboring estimates. The implementation uses `np.percentile` with the requested percentages 2.5 and 97.5.

For the $N = 300$ replicates used here, the lower bound is

$$h = 299(0.025) = 7.475, \qquad Q_{0.025} = 0.525x_7 + 0.475x_8,$$

between the eighth and ninth smallest estimates. The upper bound is

$$h = 299(0.975) = 291.525, \qquad Q_{0.975} = 0.475x_{291} + 0.525x_{292},$$

between the 292nd and 293rd smallest estimates. These bounds are taken from the ordered bootstrap distribution, rather than calculated as the fitted value plus or minus a multiple of its standard deviation.

These intervals estimate uncertainty arising from variation among the retained repeated scans, conditional on the chosen model, fit window, and fixed scan-selection masks. They do not include calibration error, uncertainty in the excluded-scan selection, unrepresented acquisition drift, or an unsuitable physical model. A narrow interval can therefore coexist with systematic residuals. The interval is also different from a conditional covariance standard error written as $\pm u$: the two calculations describe different sources of statistical variability. A 95% confidence interval is not a statement that 95% of individual measurements lie inside its bounds, or a guaranteed probability statement about this particular realized interval.

Derived quantities are recalculated for each resample. For example, $k^{*(r)} = 1/\tau^{*(r)}$ supplies a bootstrap distribution of rates. Since the reciprocal is decreasing for positive lifetimes, lifetime bounds $[L, U]$ correspond to rate bounds $[1/U, 1/L]$. The plotted rate intervals are calculated directly from the resampled rates. Neither symmetric lifetime errors nor their reciprocals should be treated as automatically symmetric rate errors.

D.3 Coefficient of determination R^2

The coefficient of determination R^2 describes how much of the variation in the fitted decay rates is accounted for by a straight line. For rates $k_i = 1/\tau_{s,i}$, their mean $\overline{k}$, and the line predictions $\widehat{k}_i = b + D_a q_i^2$, it is defined as

$$R^2 = 1 - \frac{\sum_i (k_i - \widehat{k}_i)^2}{\sum_i (k_i - \overline{k})^2}.$$

The numerator measures the squared deviations from the fitted line; the denominator measures the squared deviations from the mean rate. For the unweighted least-squares regressions with a freely fitted intercept used here, $R^2 = 1$ means that all points lie on the line, whereas $R^2 = 0$ means that the line predicts no better than the constant mean. This interpretation requires nonzero variation in the rates.

The TG-only value $R^2 = 0.934$ corresponds to 93.4% of the rate variation accounted for by the line; including PP gives $R^2 = 0.161$, or 16.1%. These are descriptive statistics with unestimated uncertainties, calculated for four and five points, respectively. Their comparison is not a formal statistical test. A high R^2 alone does not validate the physical model, quantify parameter uncertainty, or account for calibration errors and systematic residuals. It also does not show that the fitted slow components represent the same microscopic process in every dataset.

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